SHOWTIME – CASE STUDY

Project Report

Project Details:

Project Name	ShowTime – Case study
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Date	07-07-2024

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Contents

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Table of Figures

1. Project Background

An over-the-top (OTT) media service is a media service offered directly to viewers via the internet. The term is most synonymous with subscription-based video-on-demand services that offer access to film and television content, including existing series acquired from other producers, as well as original content produced specifically for the service. They are typically accessed via websites on personal computers, apps on smartphones and tablets, or televisions with integrated Smart TV platforms.

Presently, OTT services are at a relatively nascent stage and are widely accepted as a trending technology across the globe. With the increasing change in customers' social behaviour, which is shifting from traditional subscriptions to broadcasting services and OTT on-demand video and music subscriptions every year, OTT streaming is expected to grow at a very fast pace.

The global OTT market size was valued at $121.61 billion 2019 and is projected to reach $1,039.03 billion by 2027, growing at a CAGR of 29.4% from 2020 to 2027. The shift from television to OTT services for entertainment is driven by benefits such as on-demand services, ease of access, and access to better networks and digital connectivity.

With the outbreak of COVID19, OTT services are striving to meet the growing entertainment appetite of viewers, with some platforms already experiencing a 46% increase in consumption and subscriber count as viewers seek fresh content. With innovations and advanced transformations, which will enable the customers to access everything they want in a single space, OTT platforms across the world are expected to increasingly attract subscribers on a concurrent basis.

2. Objective

ShowTime is an OTT service provider that offers a wide variety of content including movies and web shows for its users. The primary reasons for the decline in viewership of content would be the decline in the number of people coming to the platform, decreased marketing spend, content timing clashes, weekends and holidays.

The primary focus will be on:

- Analysing the data provided by ShowTime to identify the primary variables driving first-day viewership, enabling proactive measures to enhance content engagement on their platform.
- Developing a linear regression model to identify the key factors influencing first-day viewership.

3. Data Description

VARIABLE	DESCRIPTION
visitors	Average number of visitors, in millions, to the platform in the past week
ad_impressions	Number of ad impressions, in millions, across all ad campaigns for the content (running and completed)
major_sports_event	Any major sports event on the day
genre	Genre of the content
dayofweek	Day of the release of the content
season	Season of the release of the content
views_trailer	Number of views, in millions, of the content trailer
views_content	Number of first-day views, in millions, of the content

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4. Exploratory Data Analysis

4.1 Data structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	visitors	ad_impressions	major_sports_event	genre	dayofweek	season	views_trailer	views_content
0	1.67	1113.81	0	Horror	Wednesday	Spring	56.70	0.51
1	1.46	1498.41	1	Thriller	Friday	Fall	52.69	0.32
2	1.47	1079.19	1	Thriller	Wednesday	Fall	48.74	0.39
3	1.85	1342.77	1	Sci-Fi	Friday	Fall	49.81	0.44
4	1.46	1498.41	0	Sci-Fi	Sunday	Winter	55.83	0.46

OBSERVATION	VARIABLES
1000	8

4.2 Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   visitors            1000 non-null   float64
 1   ad_impressions      1000 non-null   float64
 2   major_sports_event  1000 non-null   int64
 3   genre               1000 non-null   object
 4   dayofweek           1000 non-null   object
 5   season              1000 non-null   object
 6   views_trailer       1000 non-null   float64
 7   views_content       1000 non-null   float64
dtypes: float64(4), int64(1), object(3)
memory usage: 62.6+ KB
```

- The data set contains eight variables indexed from 0-7.
- The data set consists of four numerical columns of float data type and four categorical columns.
- The categorical column includes a column named 'major_sports_event' which is in int data type given in terms of 0 and 1, which indicate the absence and presence of major sports event on the day respectively.

4.3 Missing Values

- No null values were found in the data set.

4.4 Duplicate Values

- No duplicate values were found in the data set.

4.5 Statistical Summary

The following observations were made:

- **Visitors:** On average, the platform had 1.7 million visitors in the past week. A peak of 2.34 million viewers indicates the presence of outliers.
- **Ad_impressions:** The range of ad impressions spans from 1010 million to 2424 million. The difference between the number of ad impressions at the 75th percentile and the maximum value highlights the presence of outliers.
- **Genre:** There are 8 genres in the OTT platform.
- **Dayofweek:** Friday is the day when most content is released on the OTT platform.
- **Season:** During the winter season, there is a peak in visitors accessing the OTT platform.
- **Views_trailer:** The trailer has an average of 66 million views, with a maximum of 199 million views, indicating the presence of outliers.
- **Views_content:** The content has approximately half a million (0.47) first-day views, where the maximum is close to 1 million (0.89).

4.6 Count and Percentage of Categorical Variable in the Data Set

MAJOR SPORTS EVENT	COUNT	PERCENTAGE
YES	600	60
NO	400	40

GENRE	COUNT	PERCENTAGE
Others	255	25.5
Comedy	114	11.4
Thriller	113	11.3
Drama	109	10.9
Romance	105	10.5
Sci-Fi	102	10.2
Horror	101	10.1
Action	101	10.1

DAY OF WEEK	COUNT	PERCENTAGE
Friday	369	36.9
Wednesday	332	33.2
Thursday	97	9.7
Saturday	88	8.8
Sunday	67	6.7
Monday	24	2.4
Tuesday	23	2.3

SEASON	COUNT	PERCENTAGE
Winter	257	25.7
Fall	252	25.2
Spring	247	24.7
Summer	244	24.4

4.7 Univariate Analysis

4.7.1 Content Views

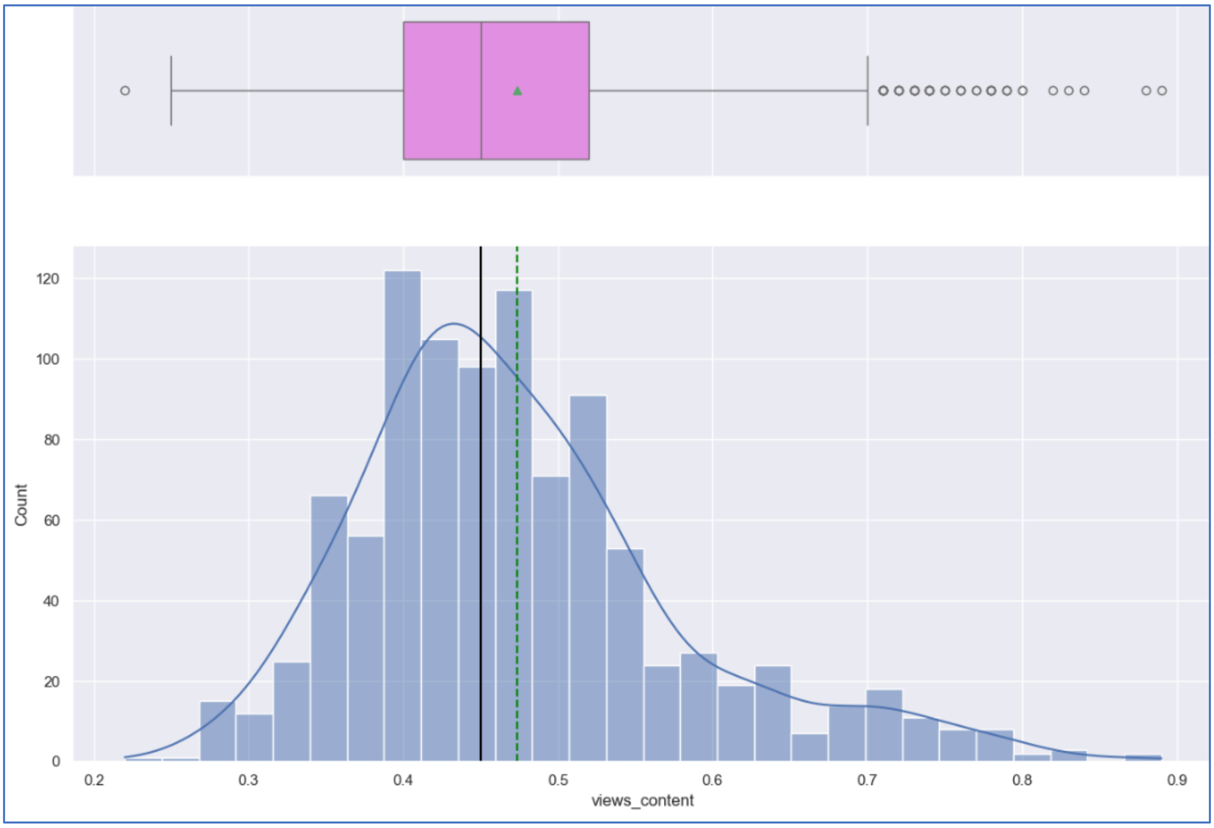


Fig 1. Content Views – Histogram and Boxplot

OBSERVATIONS:

- The content view is close to right-skewed distribution.
- The mean and median are almost close to each other, where the average being 0.49 million views on the first-day of release.
- Few of the contents are closer to 1 million views.
- The content views span over a range of 0.2 million to 1 million.

4.7.2 Trailer Views

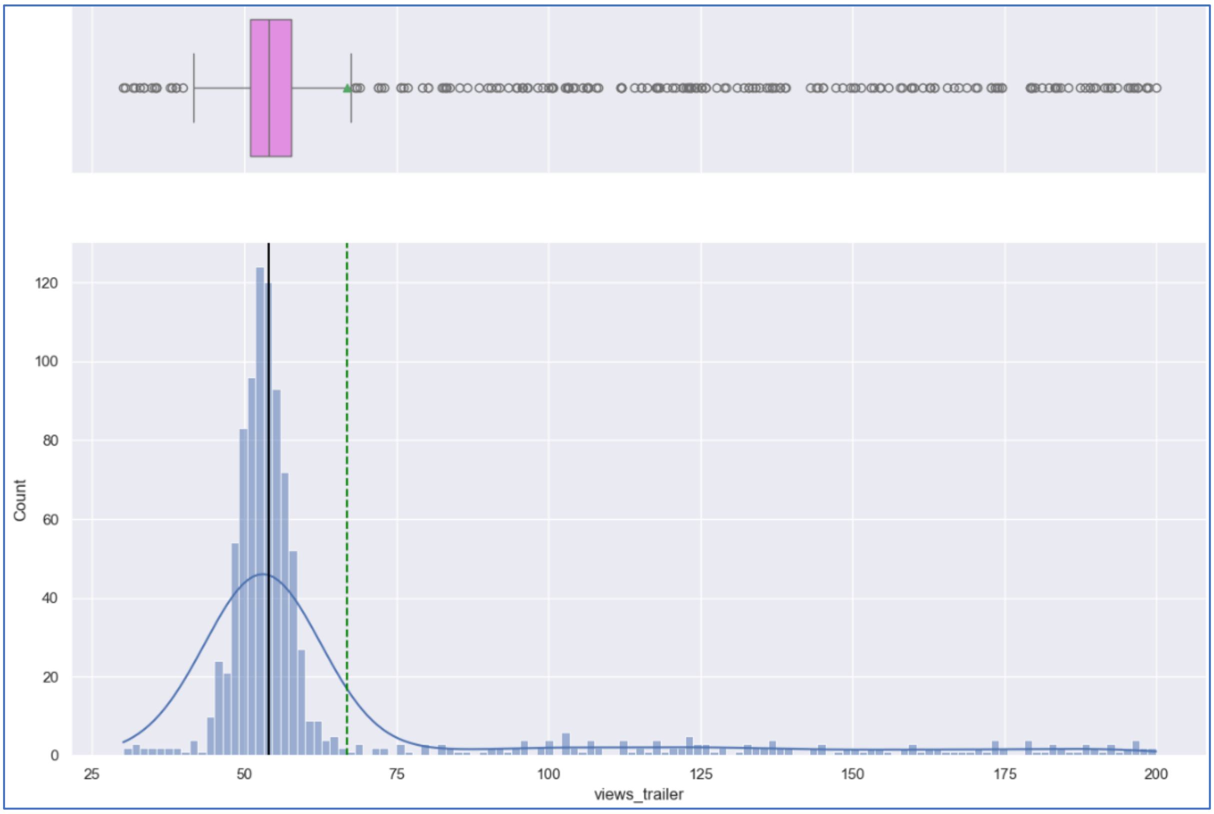


Fig 2. Trailer Views – Histogram and Boxplot

OBSERVATIONS:

- The trailer views are normally distributed with a highly right-skewed distribution.
- The average number of views are close to 65 million.
- The outliers range from 75 million to 200 million.

4.7.3 Visitors

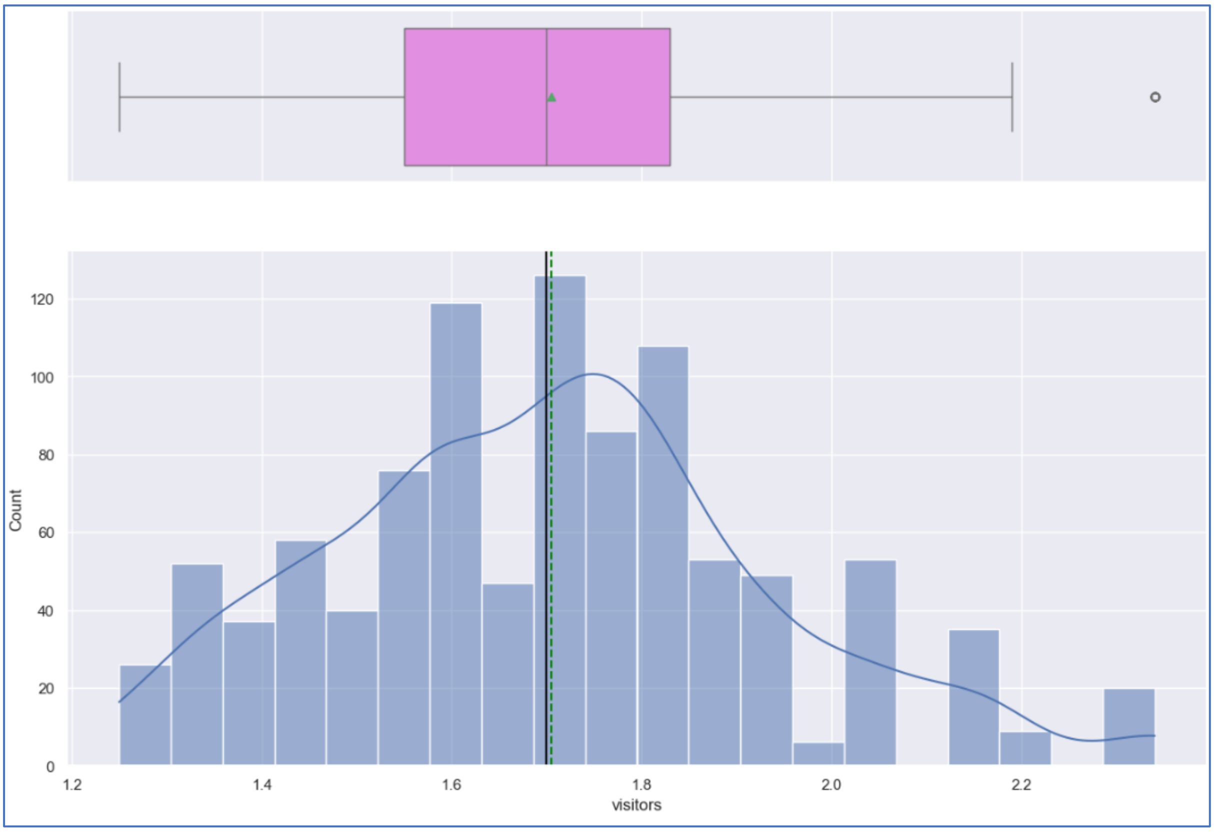


Fig 3. Visitors – Histogram and Boxplot

OBSERVATIONS:

- The mean and median are close to each other suggesting a normal distribution.
- On an average, 1.7 million people visit the platform in a week.

4.7.4 AD Impressions

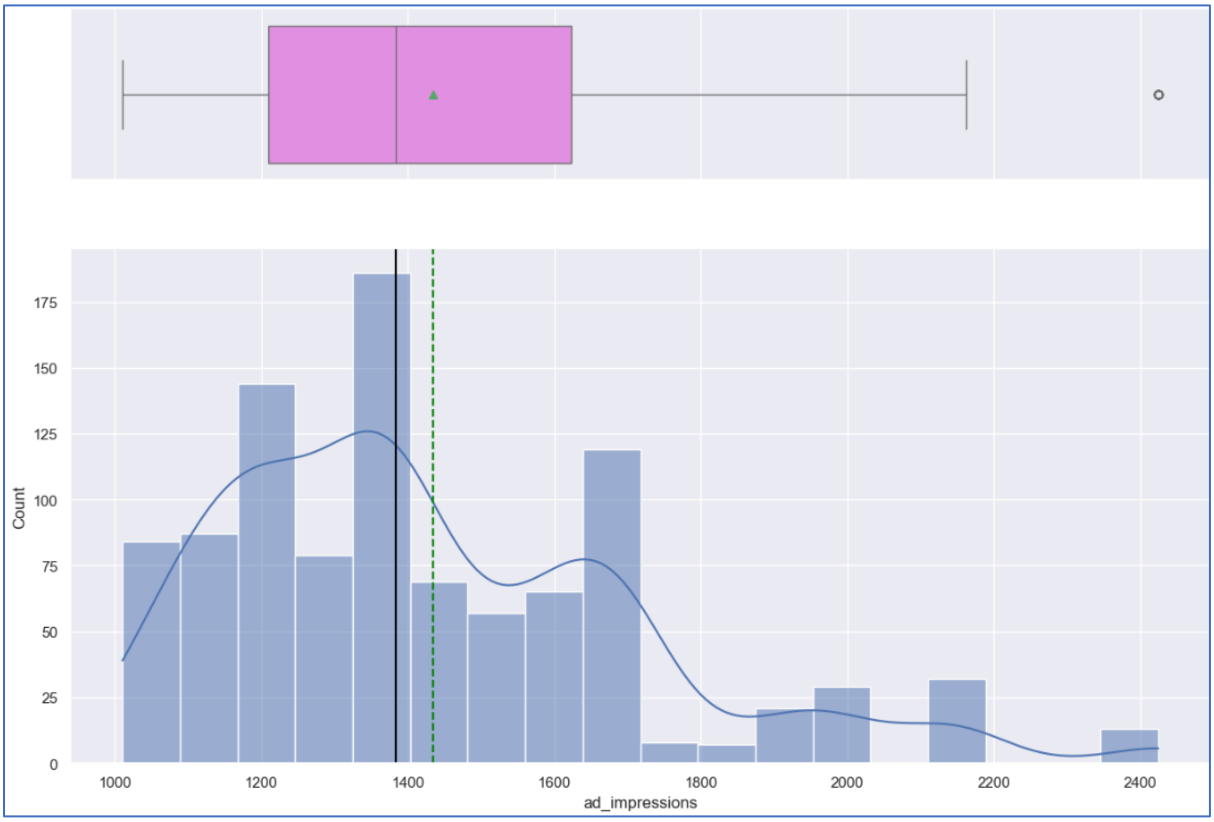


Fig 4. Ad Impressions – Histogram and Boxplot

OBSERVATIONS:

- Across all ad campaigns for the content (running and completed), the average is found to be 1450 million.
- The range of ad impressions varies from 1000 million to 2400 million.
- The distribution is found to be multimodal.

4.7.5 Major Sports Event

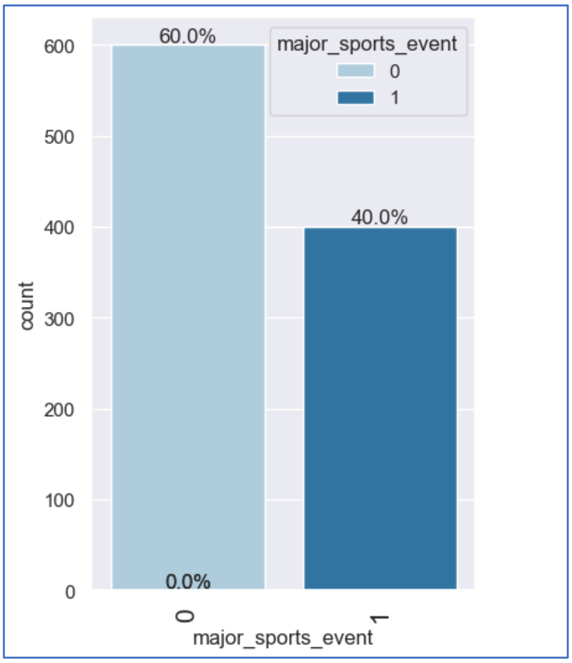


Fig 5. Major Sports Event – Bar Plot

OBSERVATIONS:

- Sports events were absent during nearly 60% of visits to the OTT platform.

4.7.6 Genre

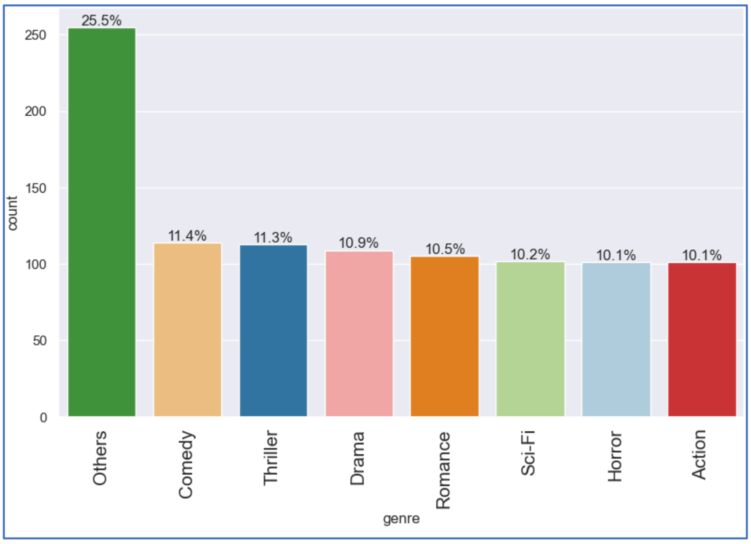


Fig 6. Genre – Bar Plot

OBSERVATIONS:

- Almost all the genres are equally distributed in the OTT platform ranging from 10.1% for action to 11.4% for comedy.
- Apart from the mentioned genres, the remaining category accounts to 25.5%.

4.7.7 Day of week

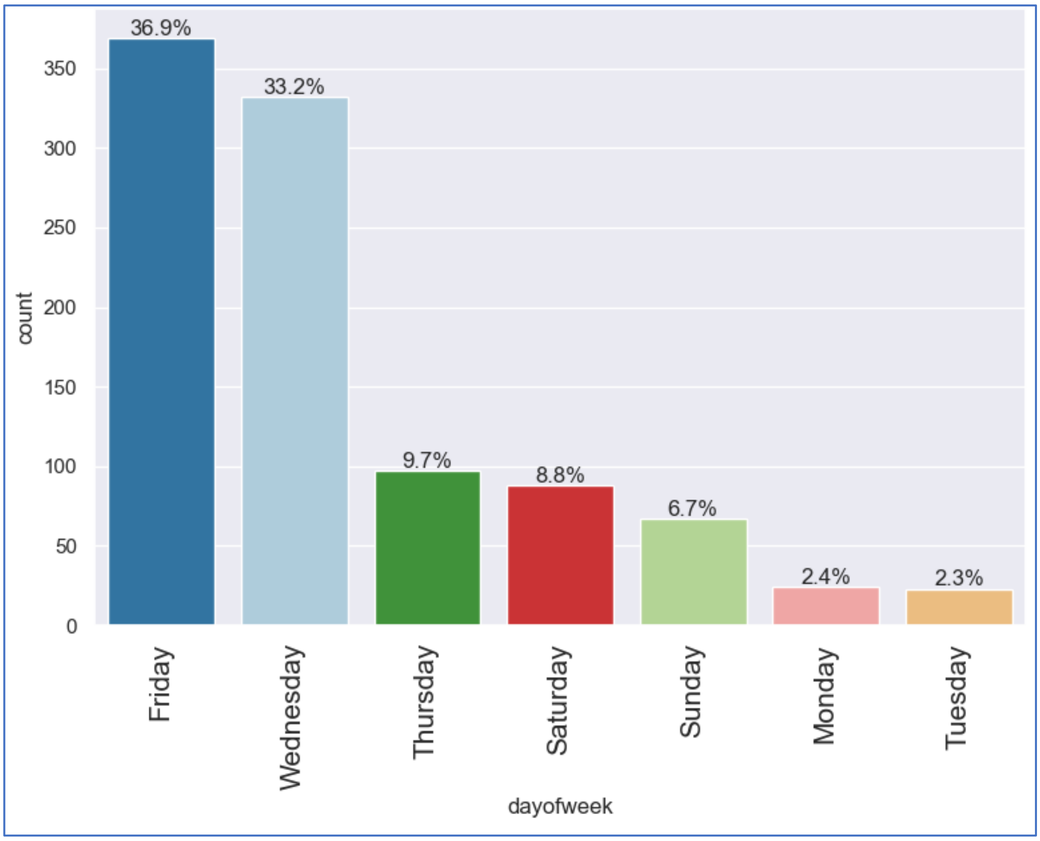


Fig 7. Day of Week – Bar Plot

OBSERVATIONS:

- Friday accounts to 36.9% when most content is released on the OTT platform.
- Wednesday ranks as the second busiest day for OTT content releases.
- Monday and Tuesday have the lowest percentages for OTT content releases, approximately 2.45% and 2.3%, respectively.

4.7.8 Season

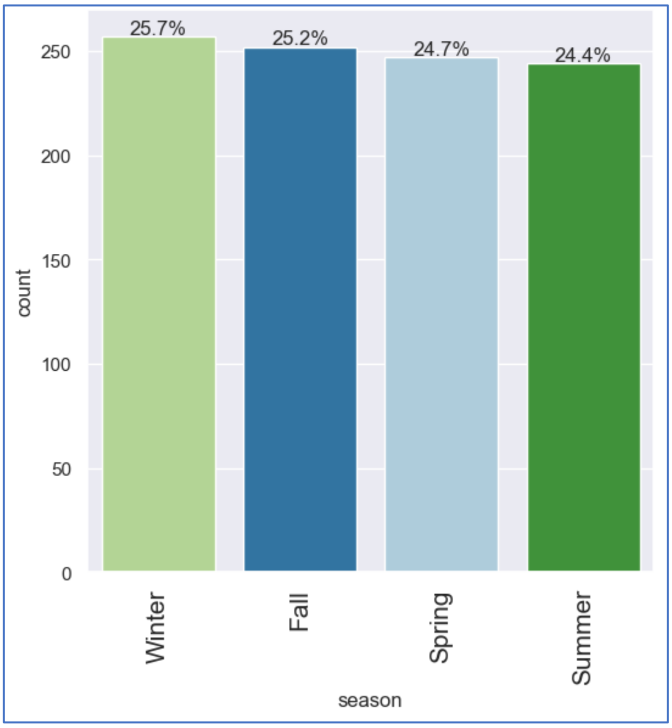


Fig 8. Season – Bar Plot

OBSERVATIONS:

- The distribution of content releases is fairly consistent across all four seasons, with winter (25.7%) slightly higher than the others which range from 24.4% to 25.2%.

4.8 Bivariate analysis

4.8.1 Heat map

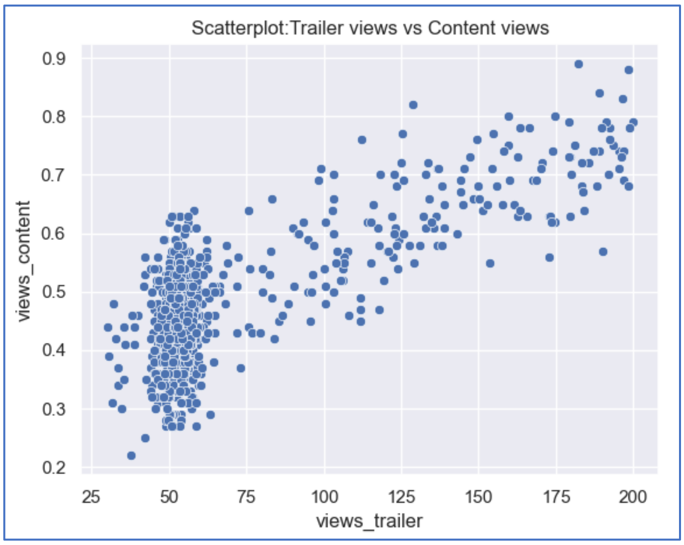
Fig 9. Heat Map

OBSERVATIONS:

- High positive correlation can be observed between number of the content trailer on the first-day and number of views of the trailer (0.75) followed by visitors visiting the OTT platform in a week and the number of views of the content trailer on the first-day (0.26).
- Negative correlation is observed between number of views of the content trailer and the visitors visiting the OTT platform in a week.
- There is no much correlation between: (a) the number of ad impressions and the visitors visiting the OTT platform in a week; (b) the number of views of the content trailer and the number of ad impressions; and (c) the number of views of the content trailer on the first-day and the number of ad impressions.
- This suggests ad impressions does not have influence on the other variables.

4.8.2 Highly correlated variables on a scatterplot

4.8.2.1 views_content v/s views_trailer



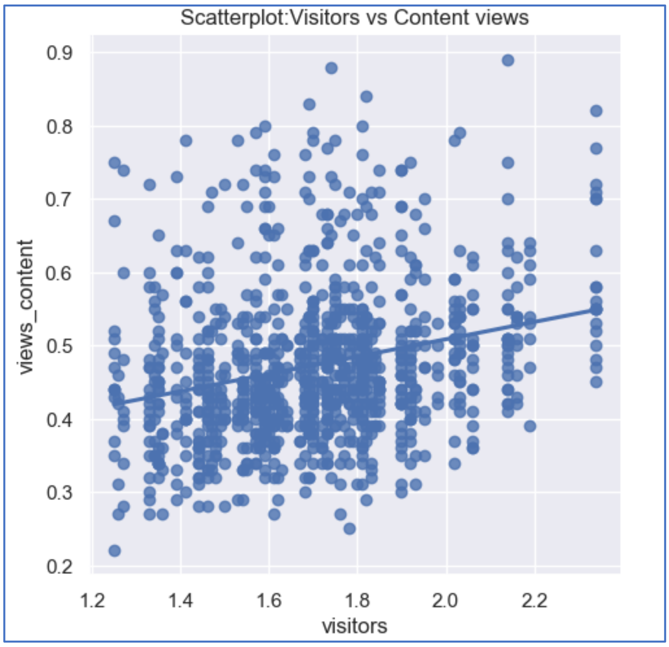
	views_trailer	views_content
views_trailer	1.000000	0.753962
views_content	0.753962	1.000000

Fig 10. views_content v/s views_trailer – Scatter Plot

OBSERVATIONS:

- As the number of views of the content trailer increases, there is increase in number of first-day views of the content.

4.8.2.2 views_content v/s visitors



	visitors	views_content
visitors	1.000000	0.259136
views_content	0.259136	1.000000

Fig 11. views_content v/s visitors – Scatter Plot

OBSERVATIONS:

- As the number of visitors to the OTT platform increases, there is a slight increase in number of first-day views of the content.

4.8.3 Pair Plot for understanding the relationship between numerical variables

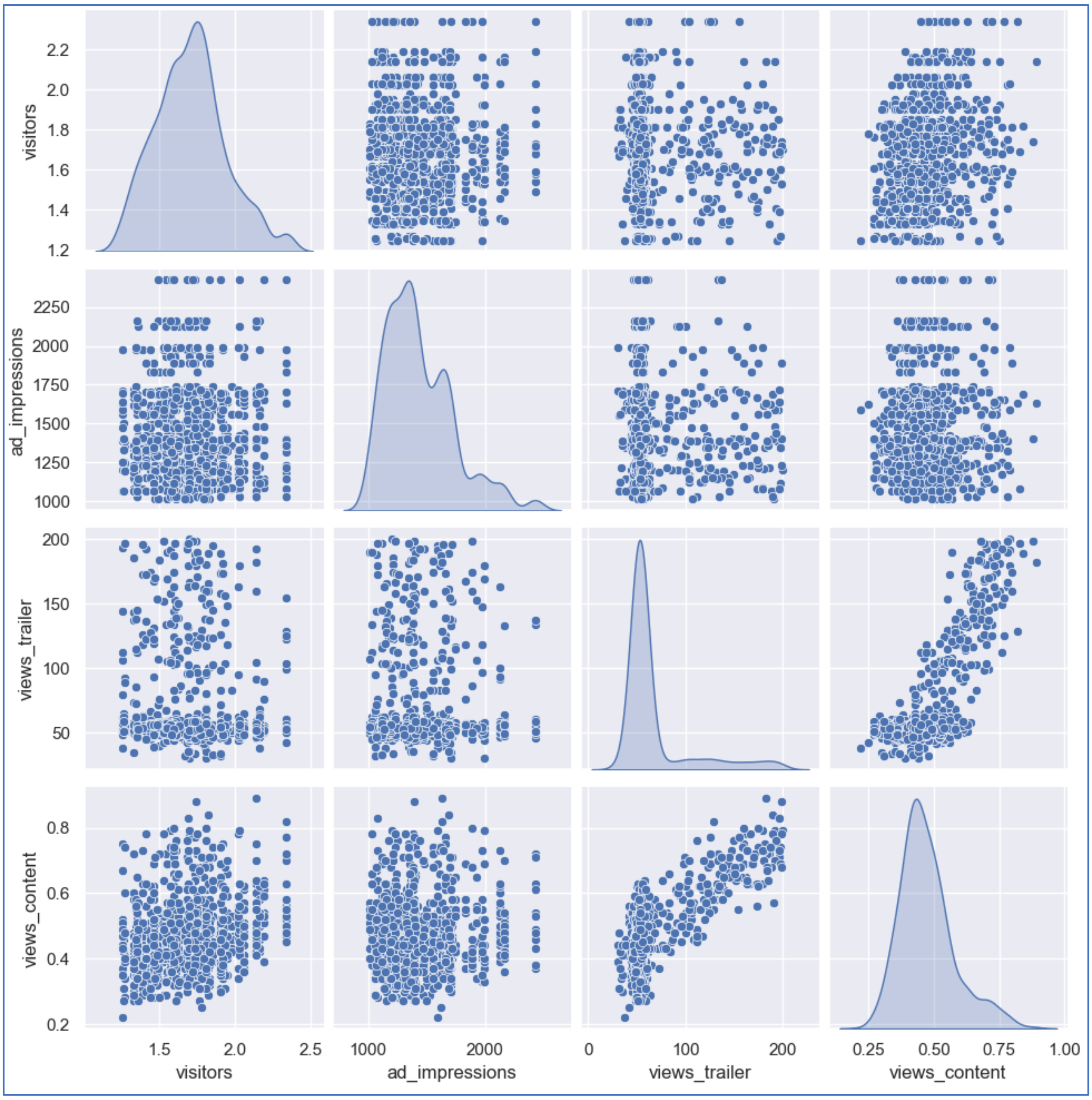


Fig 12. Numerical Variables – Scatter Plot

OBSERVATIONS:

- The first-day views of the content are influenced linearly by both the number of views for the content trailer and the number of visitors visiting the OTT platform.

4.8.4 views_content v/s views_trailer related to categorical variables (genre, season, day of week and major sports event)

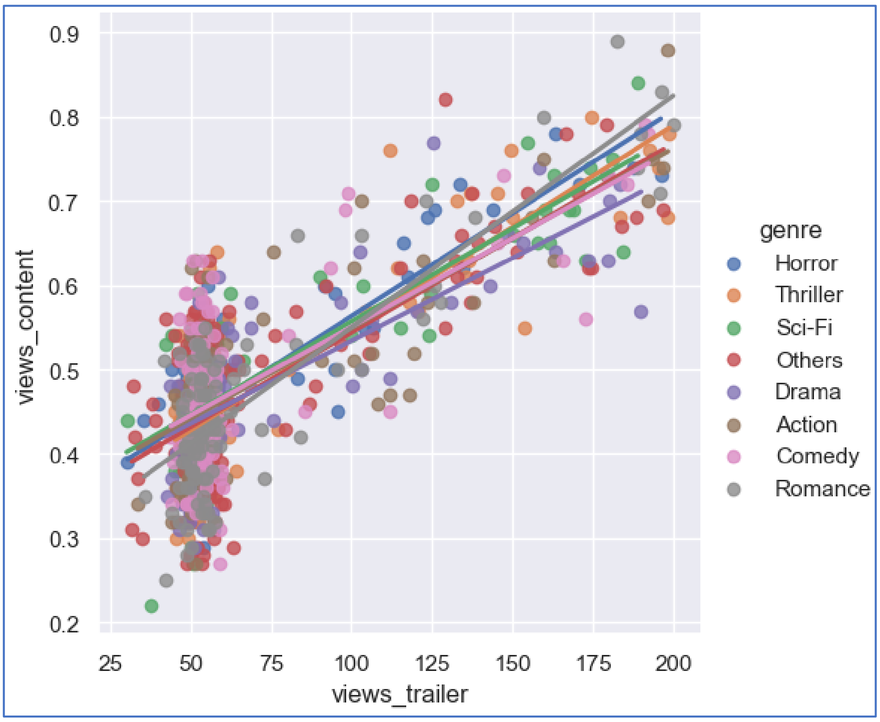


Fig 13. view_content v/s views_trailer (genre)– Lm Plot

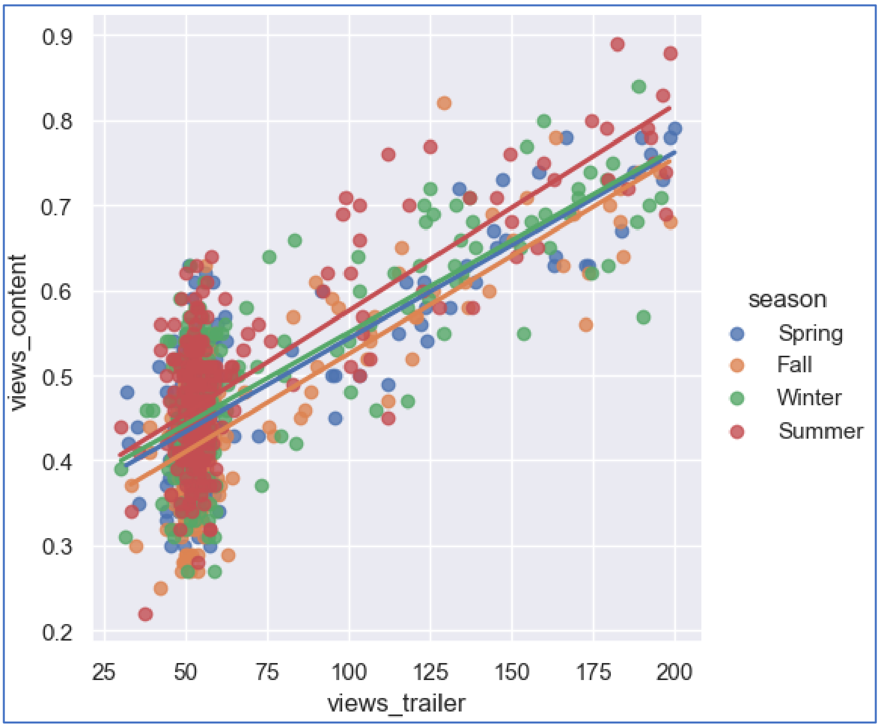


Fig 14. views_content v/s views_trailer (season)– Lm Plot

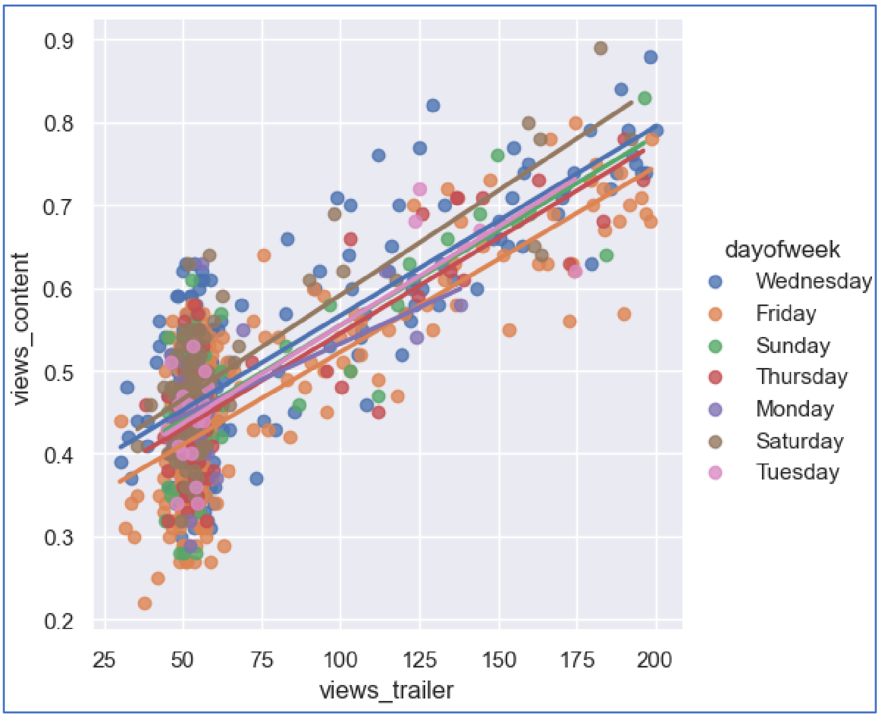


Fig 15. views_content v/s views_trailer (dayofweek)– Lm Plot

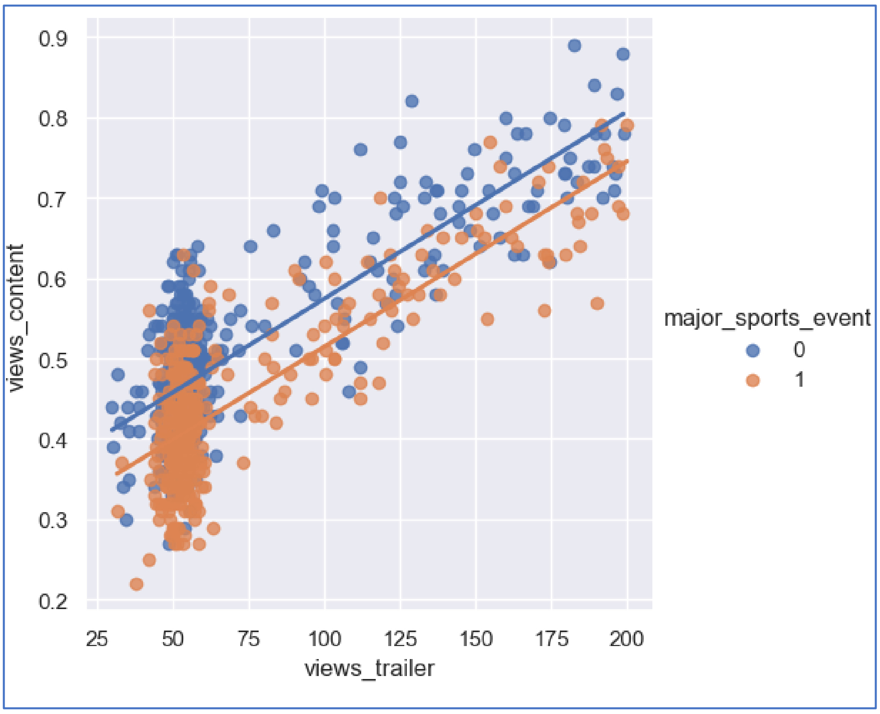


Fig 16. views_content v/s views_trailer (major_sports_event)– Lm Plot

OBSERVATIONS:

- The number of first-day views of the content demonstrates a linear relationship with the number of views of the content trailer across all genres, seasons, days of the week, and irrespective of whether a sports event occurred.

4.8.5 views_content v/s visitors related to categorical variables (genre, season, day of week and major sports event)

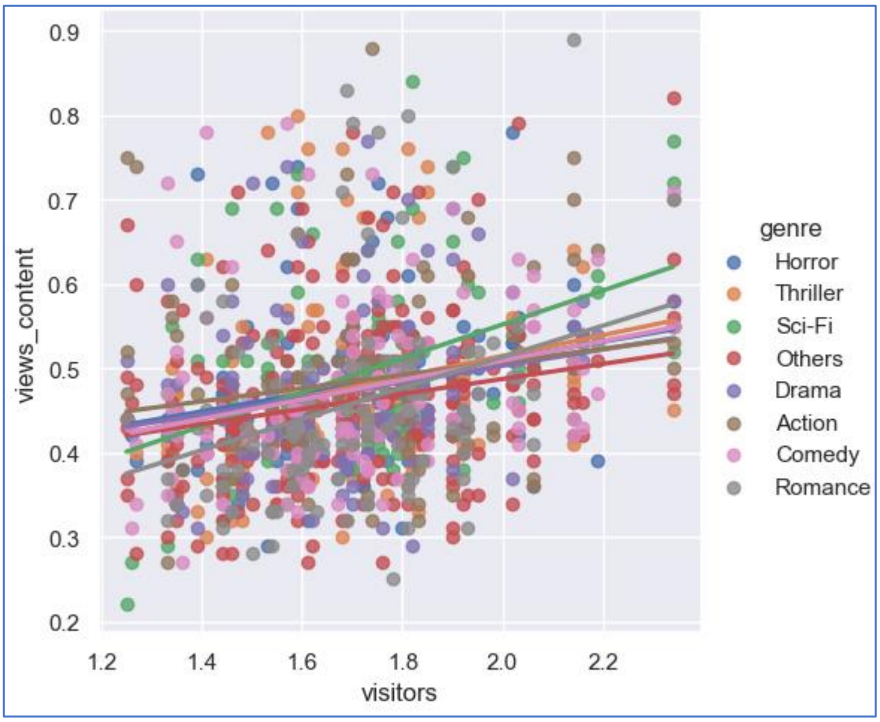


Fig 17. views_content v/s visitors (genre)– Lm Plot

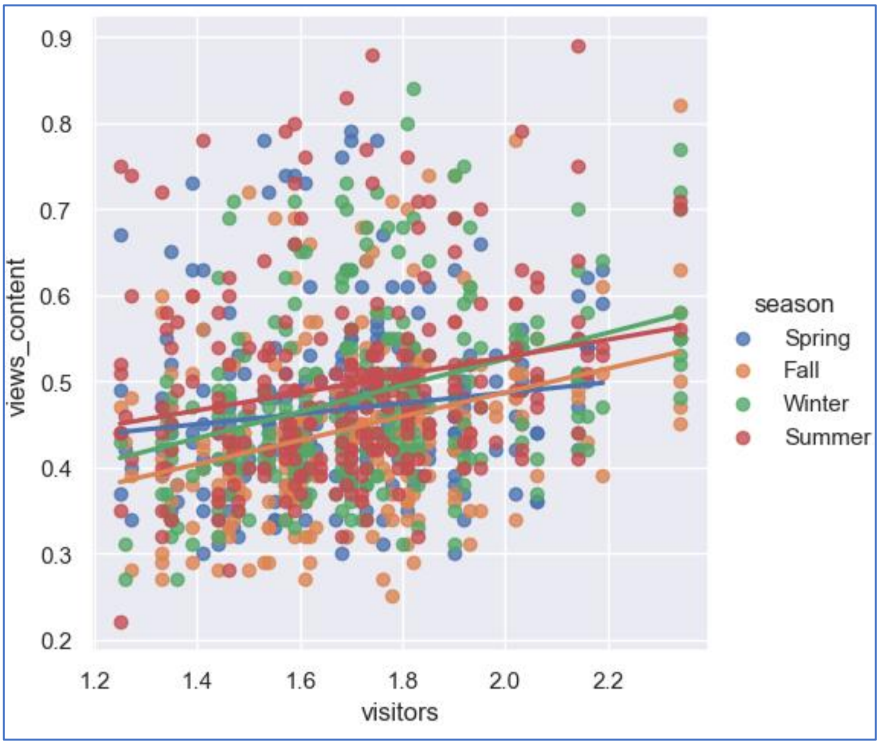


Fig 18. views_content v/s visitors (season)– Lm Plot

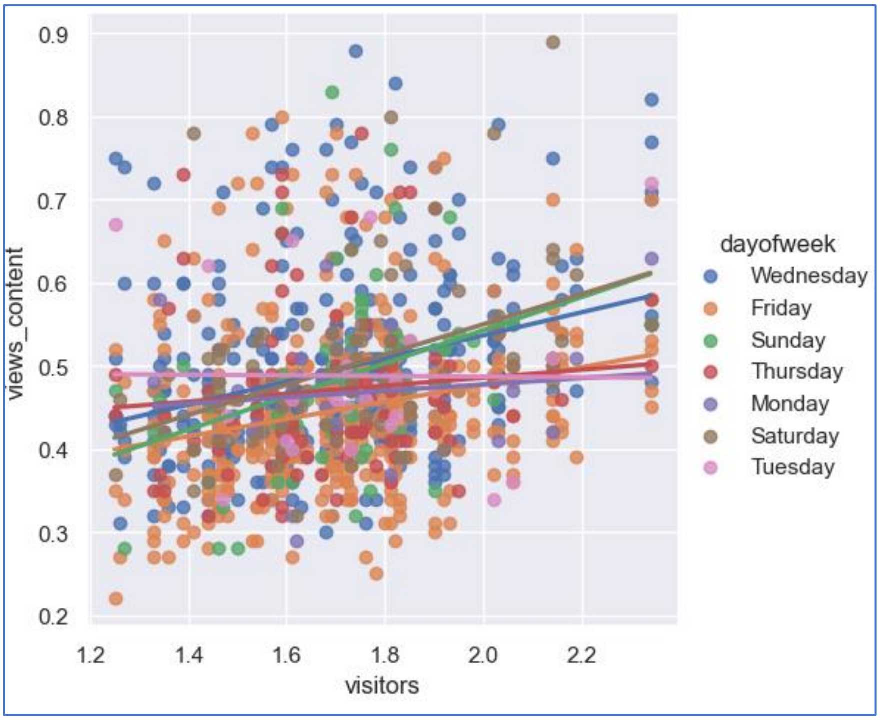


Fig 19. views_content v/s visitors (dayofweek)– Lm Plot

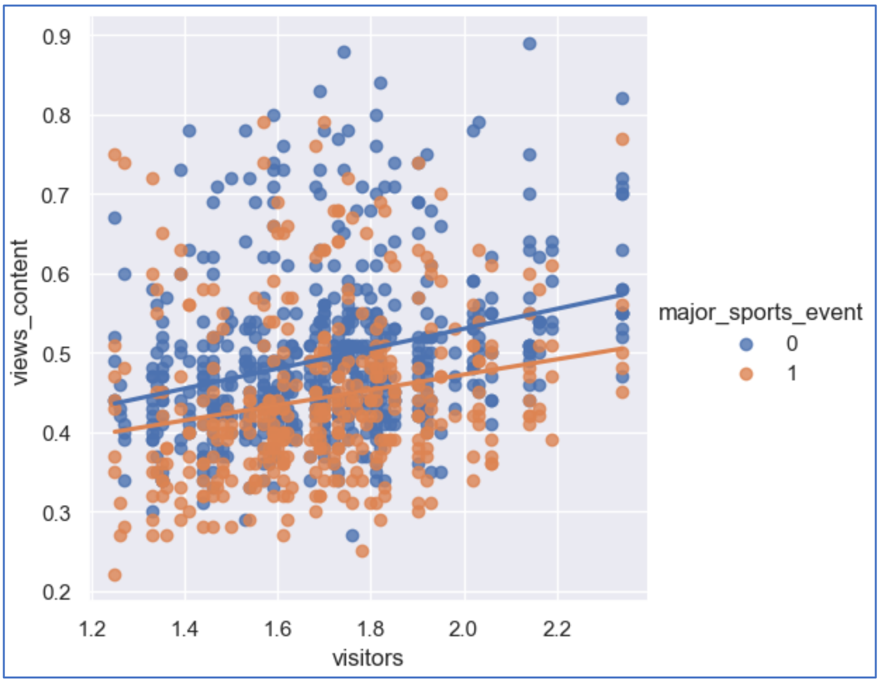


Fig 20. views_content v/s visitors (major_sports_event)– Lm Plot

OBSERVATIONS:

- The number of first-day views of the content demonstrates a linear relationship with the number of visitors across all genres, seasons, days of the week, and irrespective of whether a sports event occurred.

4.8.6 Relationship between views_content and categorical columns in the data

4.8.6.1 views_content V/S major_sports_event

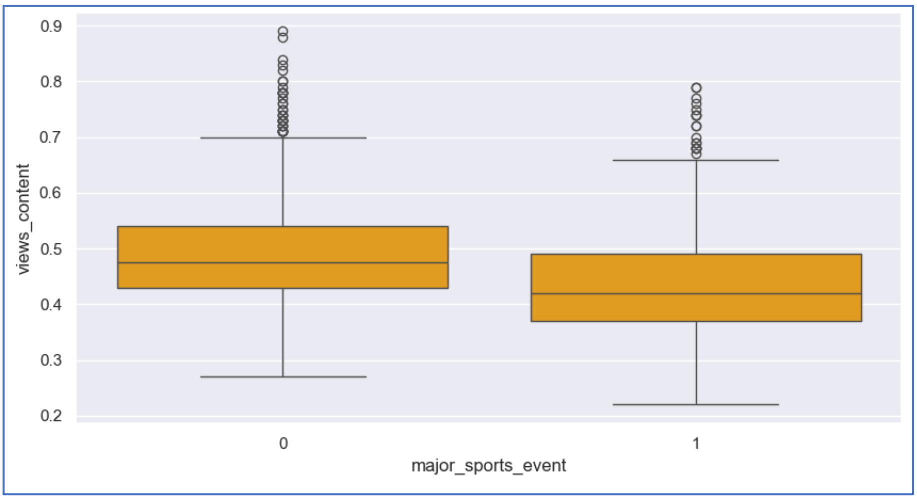


Fig 21. views_content v/s major_sports_event – Box Plot

OBSERVATIONS:

- On an average, the absence of sports event shows higher number of views, in millions, on the first-day of content release, which accounts approximately close to 0.5 million.

4.8.6.2 views_content V/S dayofweek

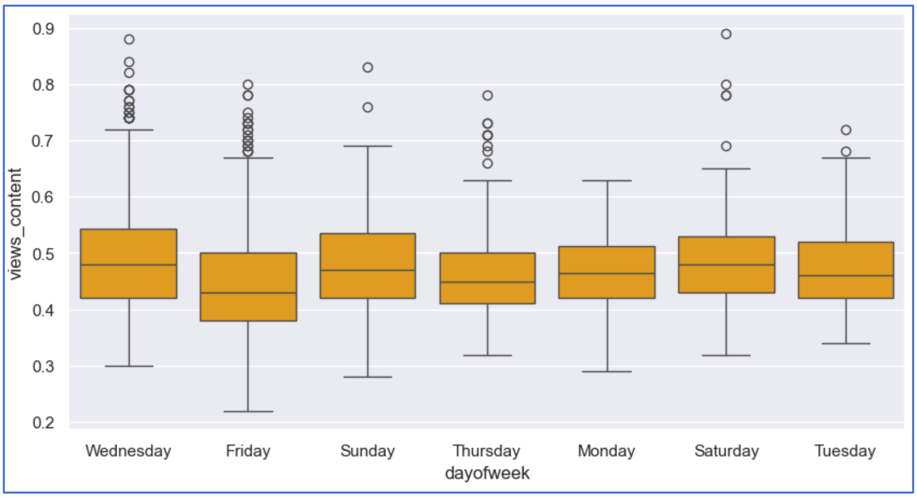


Fig 22. views_content v/s dayofweek – Box Plot

OBSERVATIONS:

- The contents released on Wednesdays, Saturdays and Sundays, have high number of views on the first-day of content release.
- The contents released on Mondays and Thursdays have low number of views, in millions, on the first-day of content release comparatively.

4.8.6.3 views_content V/S season

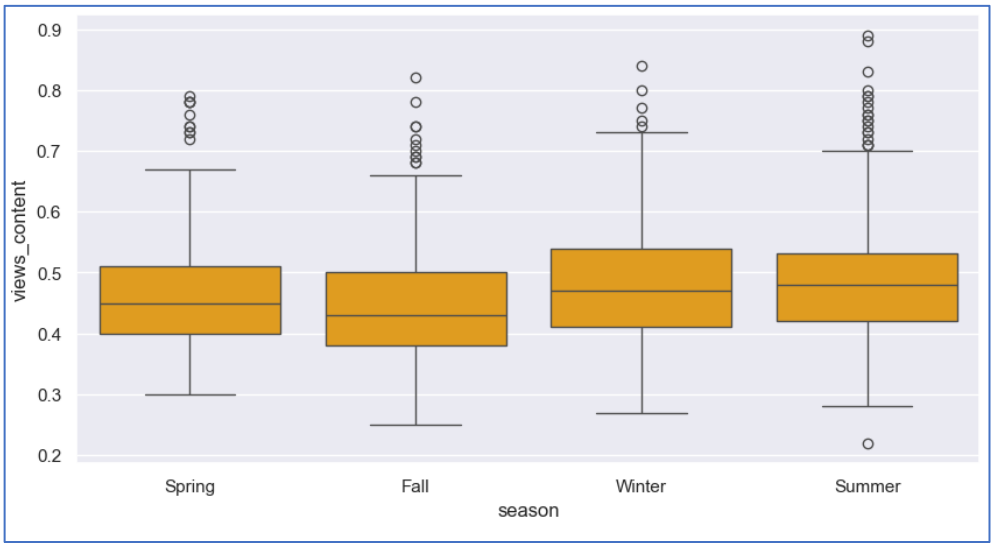


Fig 23. views_content v/s season – Box Plot

OBSERVATIONS:

- Content released during Summer receives a significant number of views on its first day of release, followed by Winter.
- Content released in Spring and Fall receive relatively fewer views on their first day of release.

4.8.6.4 views_content V/S genre

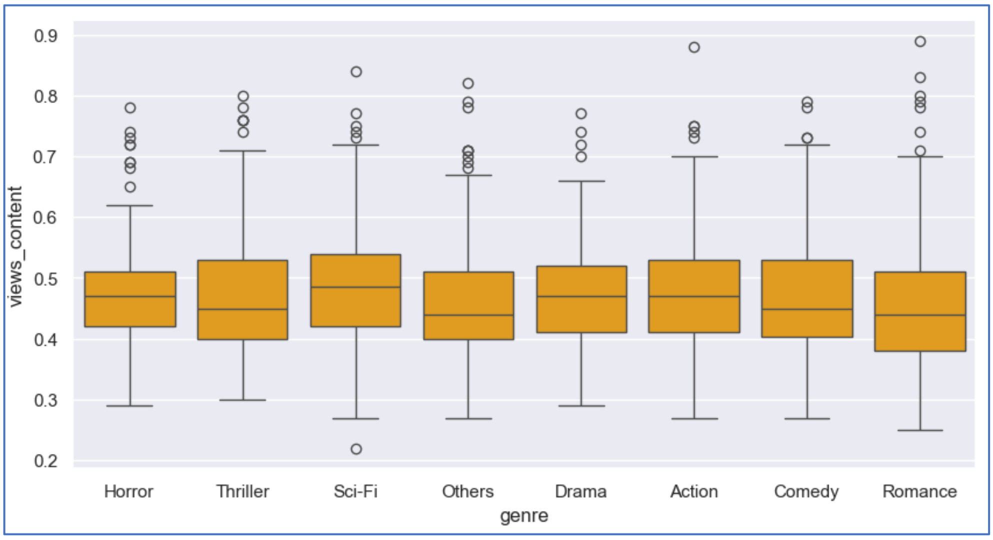


Fig 24. Views_content v/s genre – Box Plot

OBSERVATIONS:

- Sci-Fi, Horror and Action contents have highest number of views on the first day of release.
- Drama and Romance are least watched on the first day of content release.

4.8.7 Relationship between visitors on the OTT platform and categorical columns in the data

4.8.7.1 visitors V/S major_sports_event

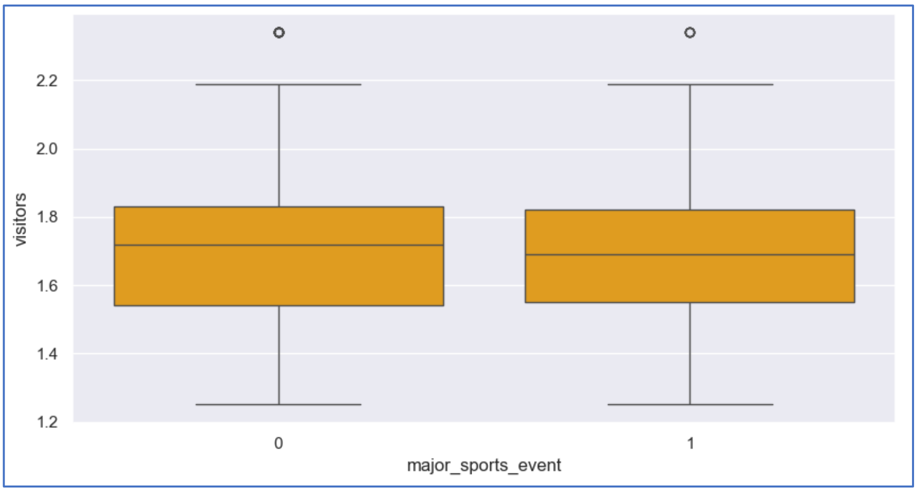


Fig 25. visitors v/s major_sports_event – Box Plot

OBSERVATIONS:

- The mean number of visitors is slightly higher when there is no major sports event compared to when there is a major sports event.

4.8.7.2 visitors V/S dayofweek

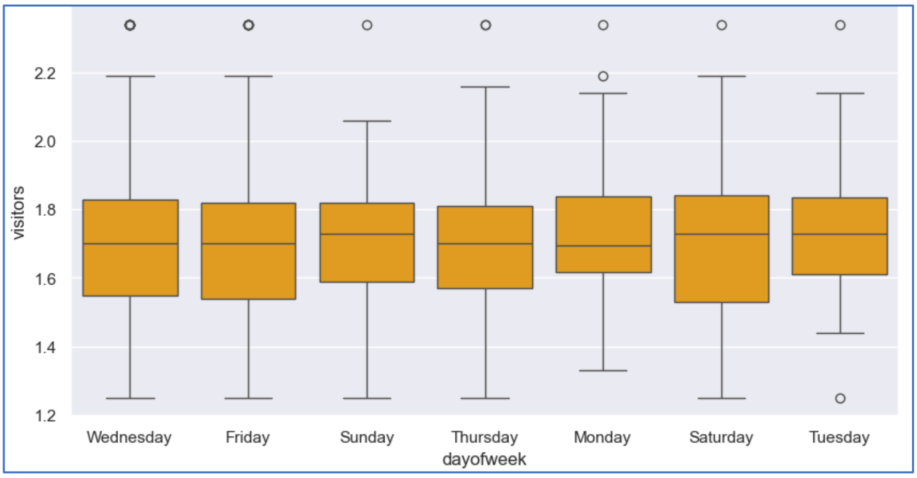


Fig 26. visitors v/s dayofweek – Box Plot

OBSERVATIONS:

- On an average, the visitors visiting the OTT platform are seen to be higher on contents released on Sunday, Saturday and Tuesday.

4.8.7.3 visitors V/S season

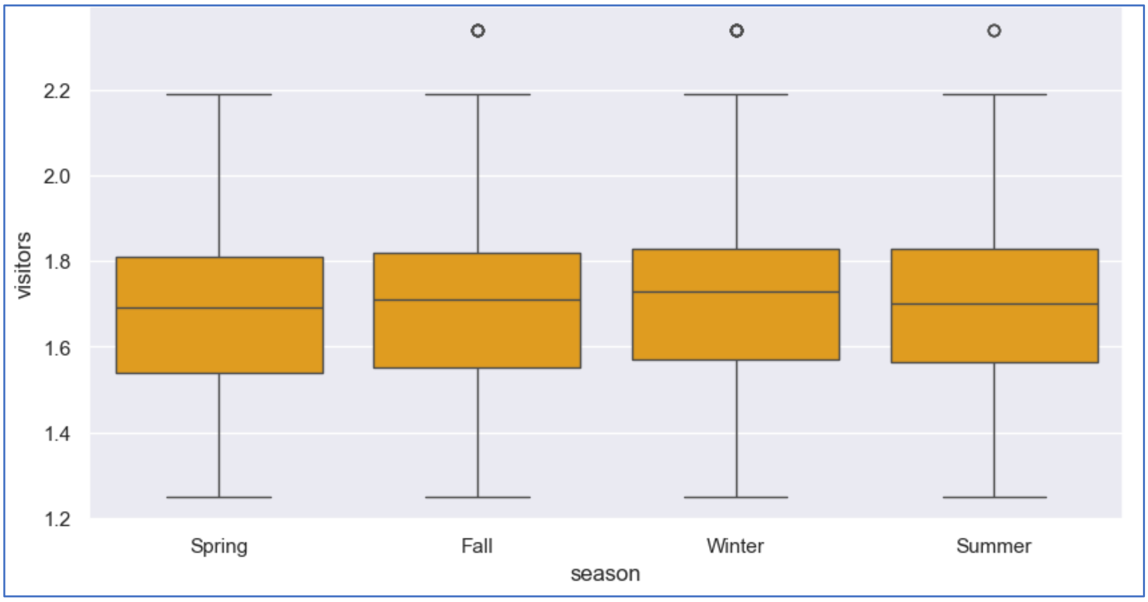


Fig 27. visitors v/s season – Box Plot

OBSERVATIONS:

- Visitor traffic on the OTT platform tends to peak during Winter and decrease notably during Spring.

4.8.7.4 visitors v/s genre

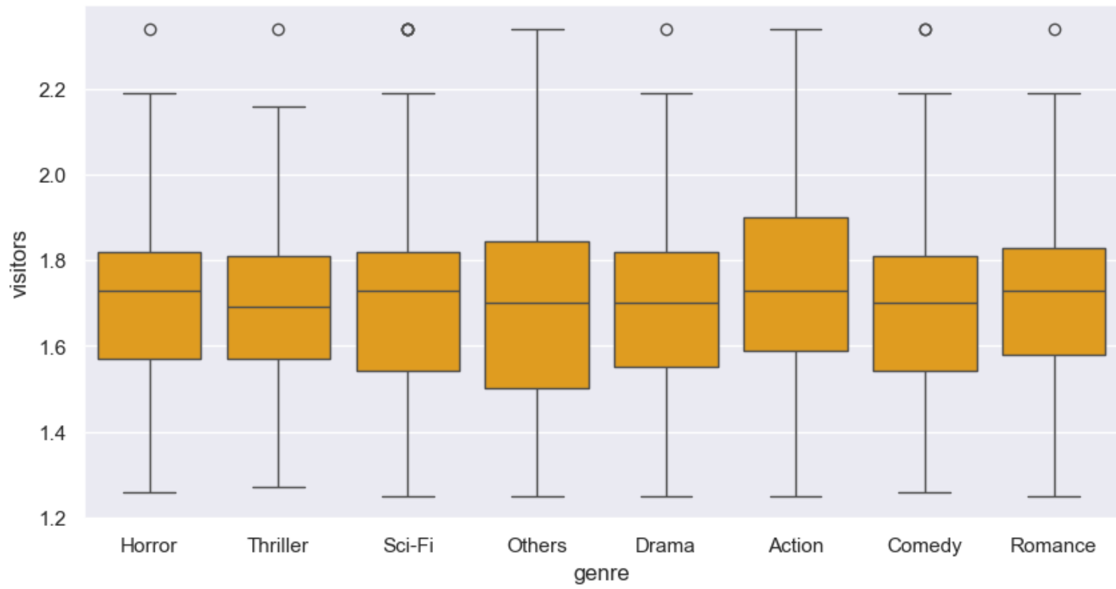


Fig 28. visitors v/s genre – Box Plot

OBSERVATIONS:

- On average, Sci-Fi, Action, Romance and Horror genres attract significant viewership on the OTT platform.

4.8.8 Relationship between ad impressions on the OTT platform and categorical columns in the data

4.8.8.1 ad_impressions V/S major_sports_event

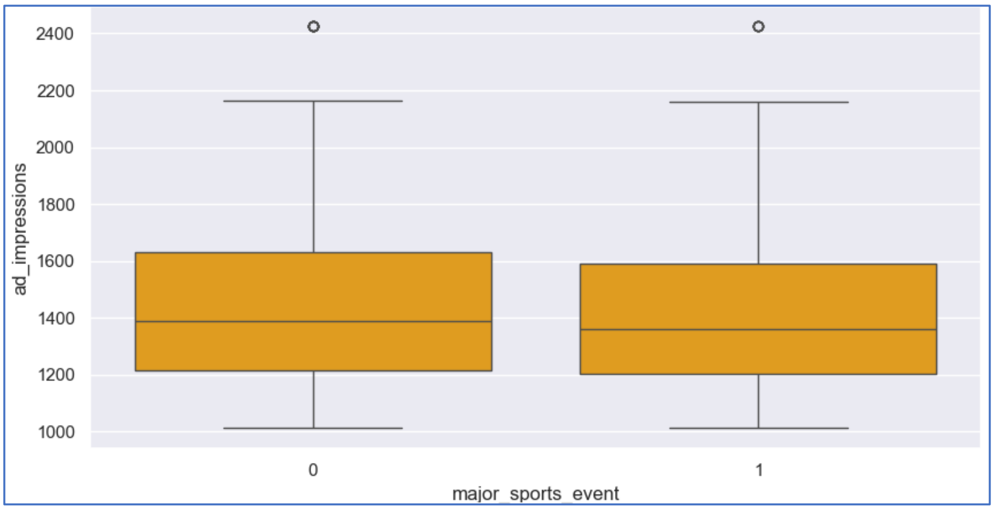


Fig 29. ad_impressions v/s major_sports_event – Box Plot

OBSERVATIONS:

- Ad impressions tend to be slightly higher when there are no major sports events compared to periods when such events are taking place.

4.8.8.2 ad_impressions V/S dayofweek

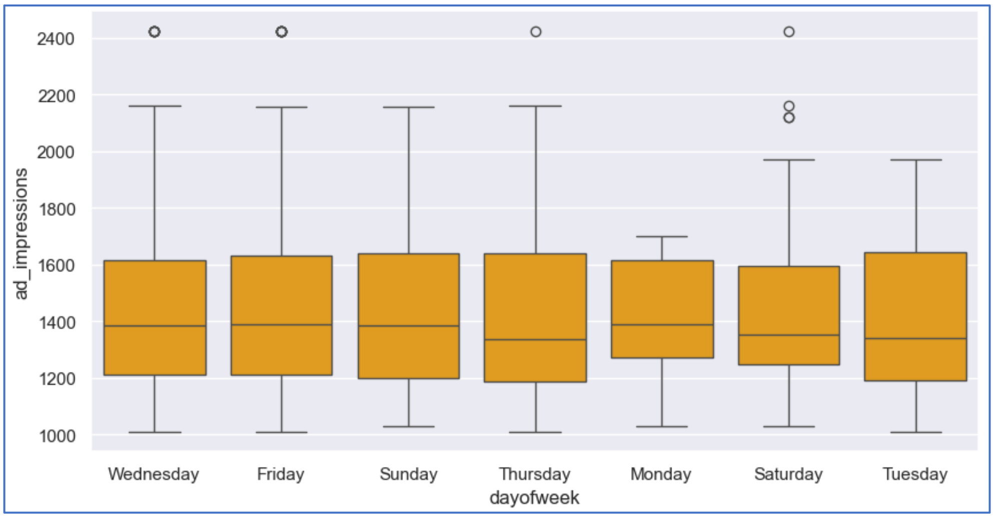


Fig 30. ad_impressions v/s dayofweek – Box Plot

OBSERVATIONS:

- Ad impressions peak on content released on Fridays, Sundays, and Mondays, while they decline on Thursdays.

4.8.8.3 ad_impressions V/S season

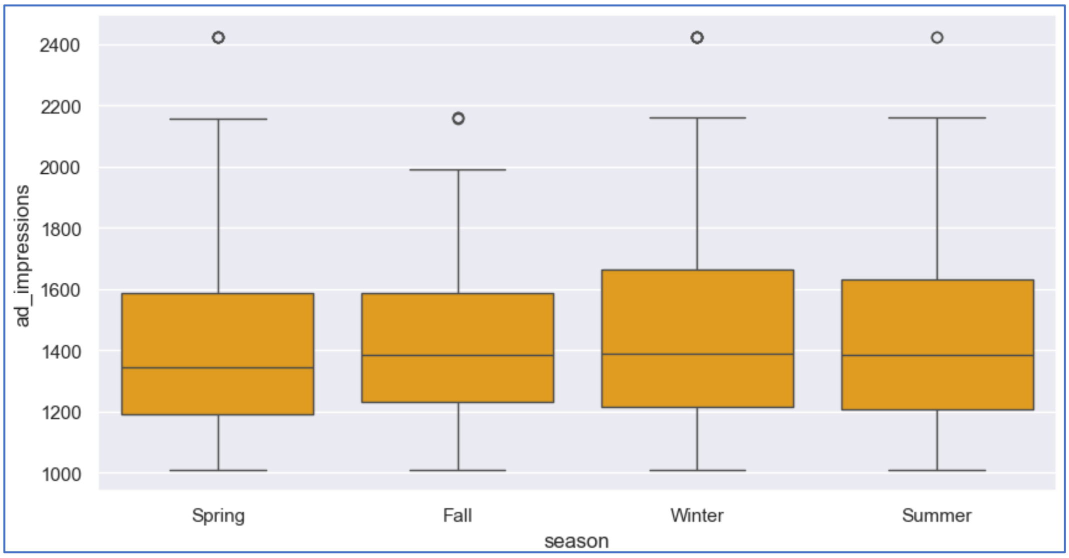


Fig 31. ad_impressions v/s season – Box Plot

OBSERVATIONS:

- Ad impressions reach their peak on content released in Winter and significantly decline in Spring.

4.8.8.4 ad_impressions V/S genre

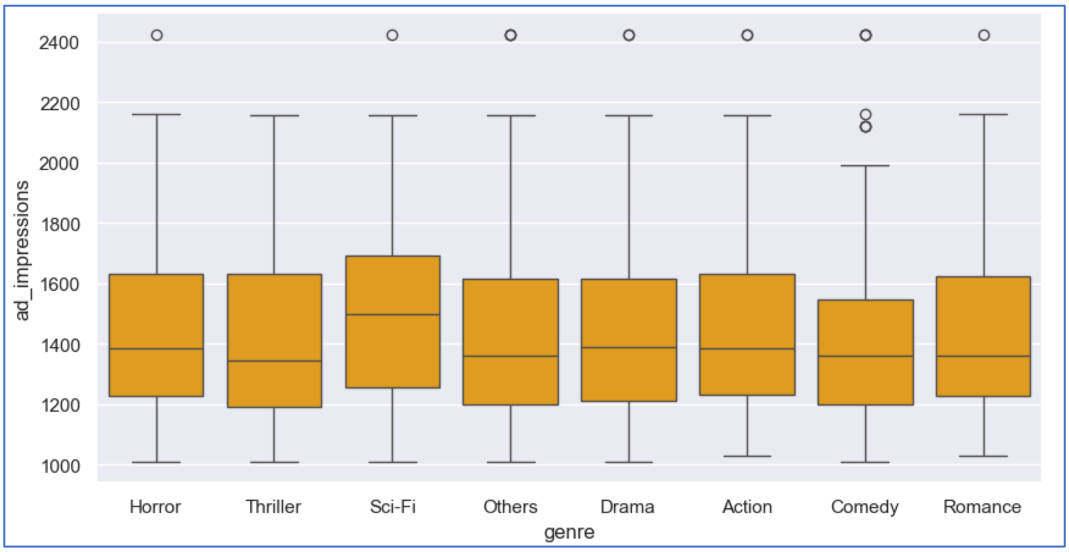


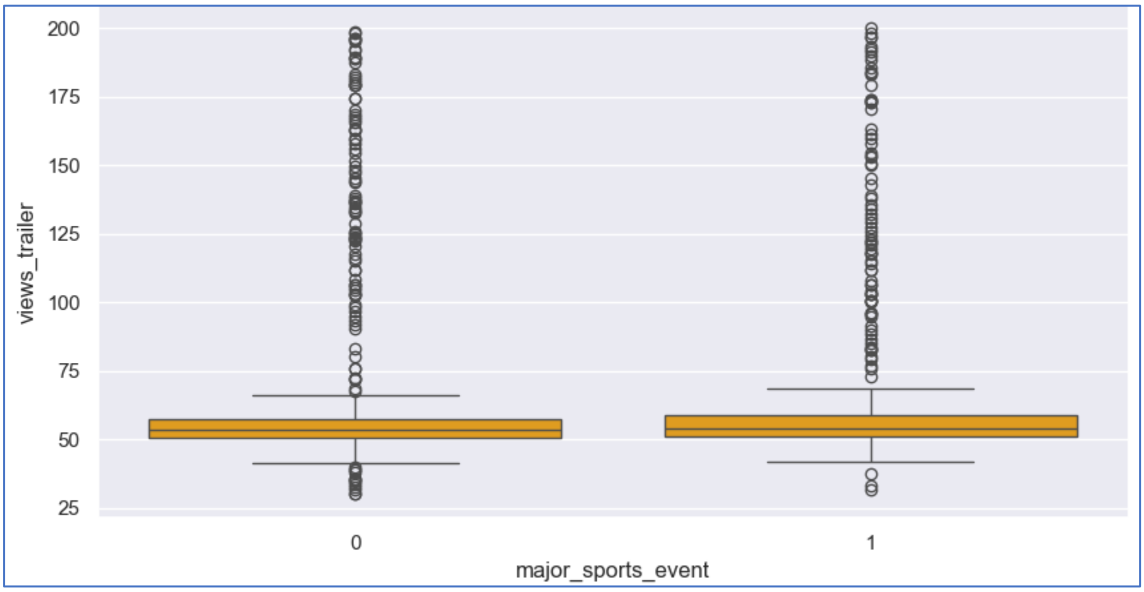
Fig 32. ad_impressions v/s genre – Box Plot

OBSERVATIONS:

- Ad impressions are highest for Sci-Fi and Drama compared to all other genres.

4.8.9 Relationship between trailer views on the OTT platform and categorical columns in the data

4.8.9.1 views_trailer V/S major_sports_event



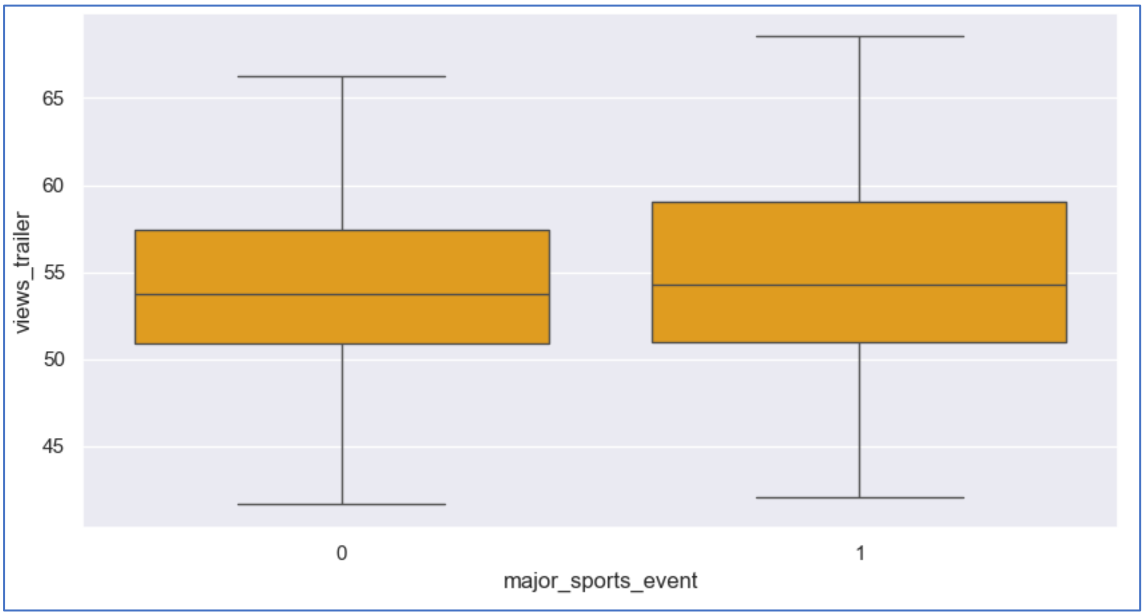
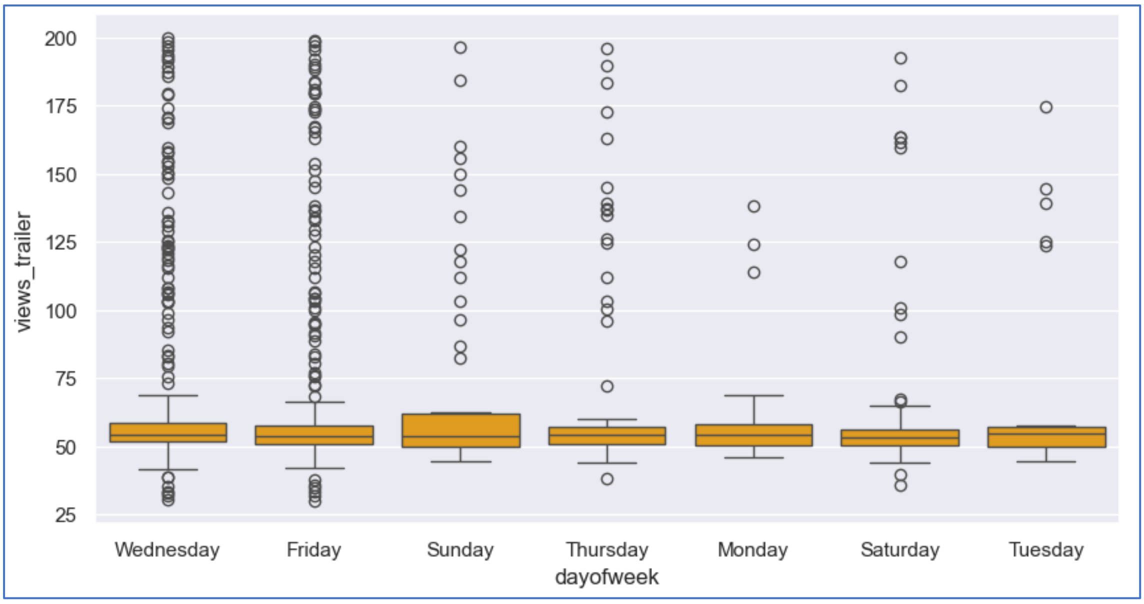


Fig 33. views_trailer v/s major_sports_event – Box Plot

OBSERVATIONS:

- Content trailer views experience a slight increase during major sports events.

4.8.9.2 views_trailer V/S dayofweek



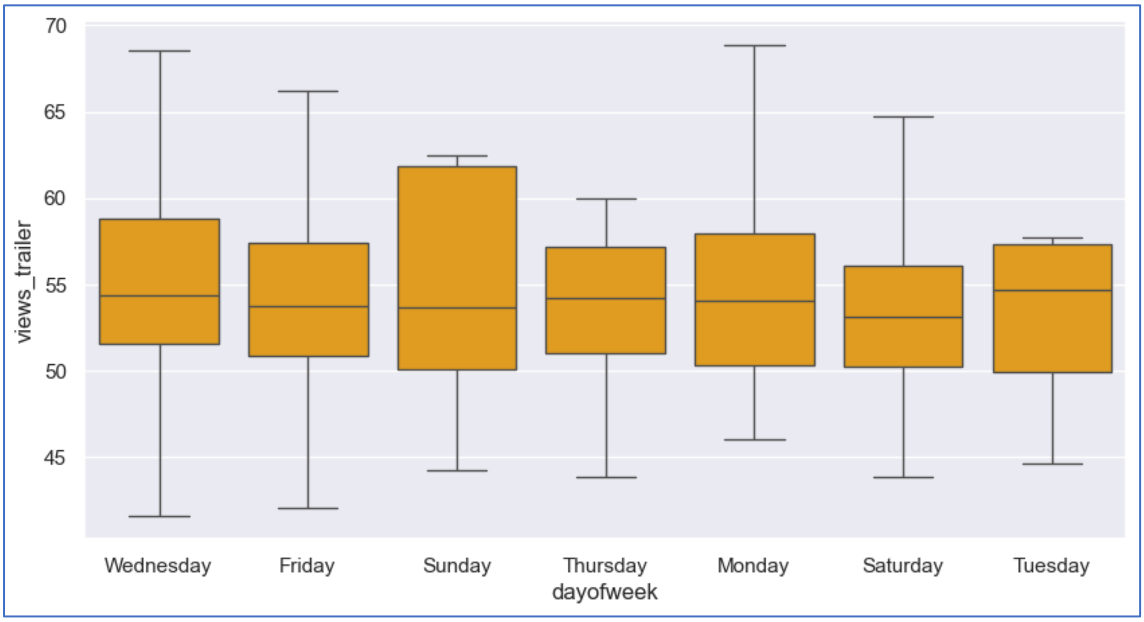
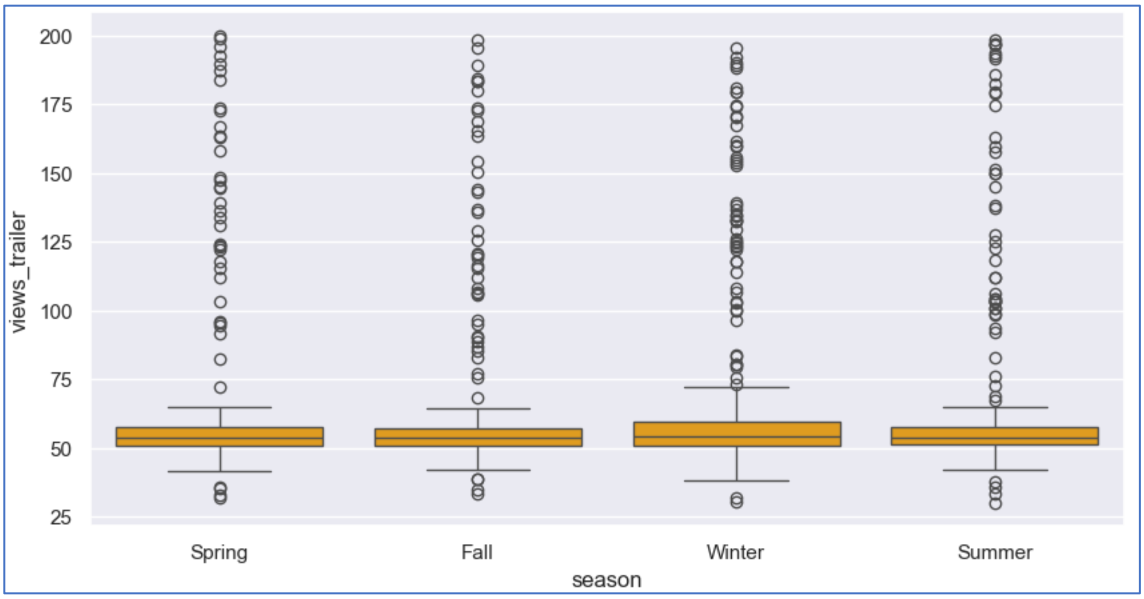


Fig 34. views_trailer v/s dayofweek – Box Plot

OBSERVATIONS:

- On average, content trailer views see a slight increase for content released on Tuesdays, Wednesdays, and Thursdays.

4.8.9.3 views_trailer v/s season



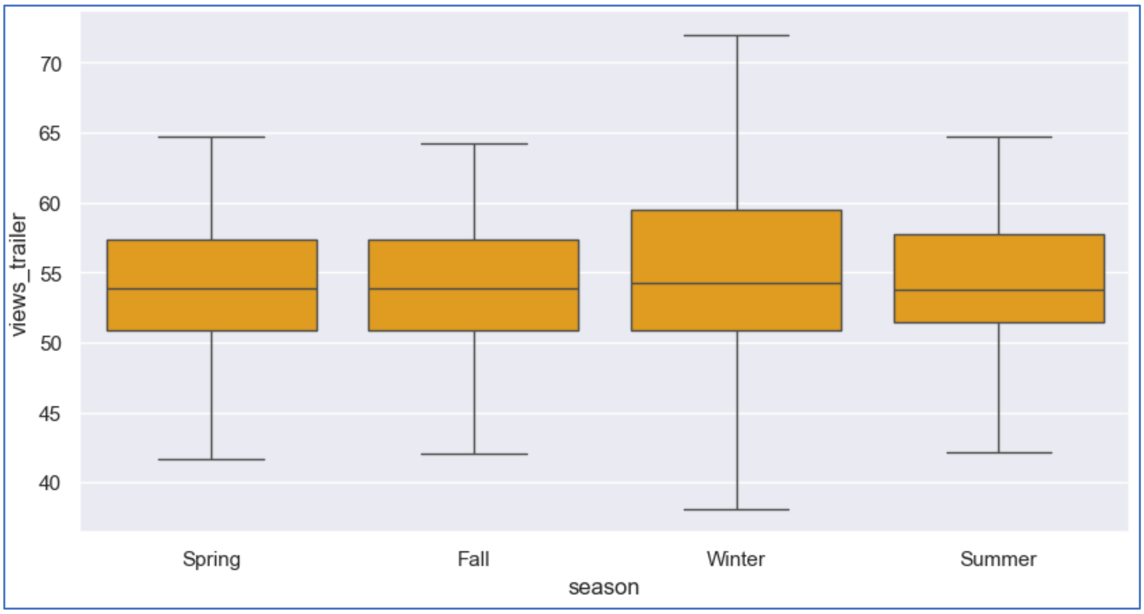
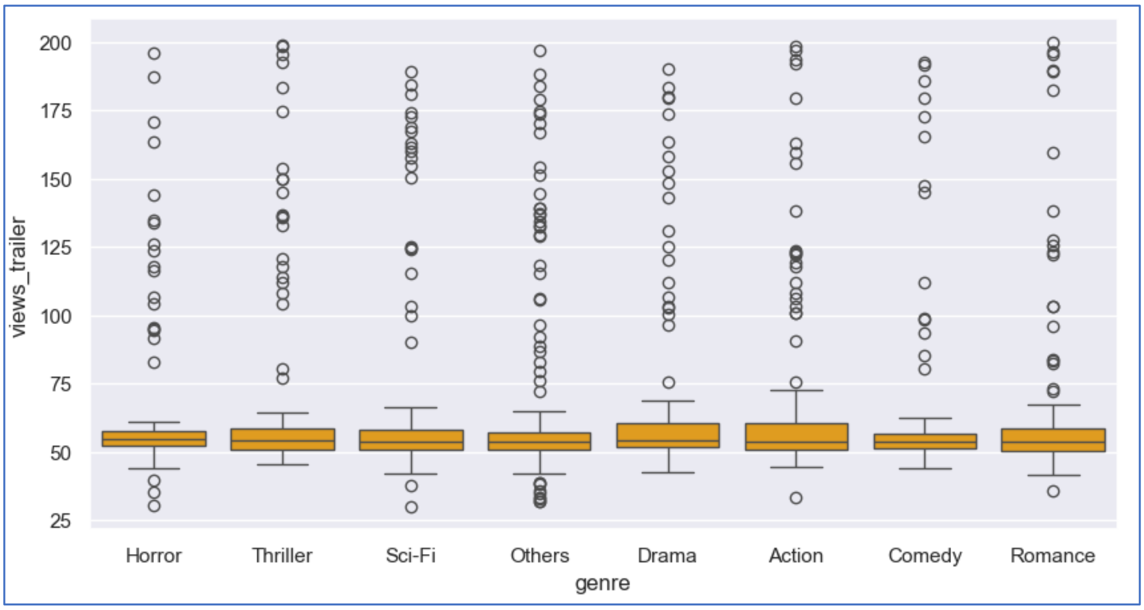


Fig 35. views_trailer v/s season – Box Plot

OBSERVATIONS:

- Content trailers experience slightly higher viewership for content release in Winter.

4.8.9.4 views_trailer v/s genre



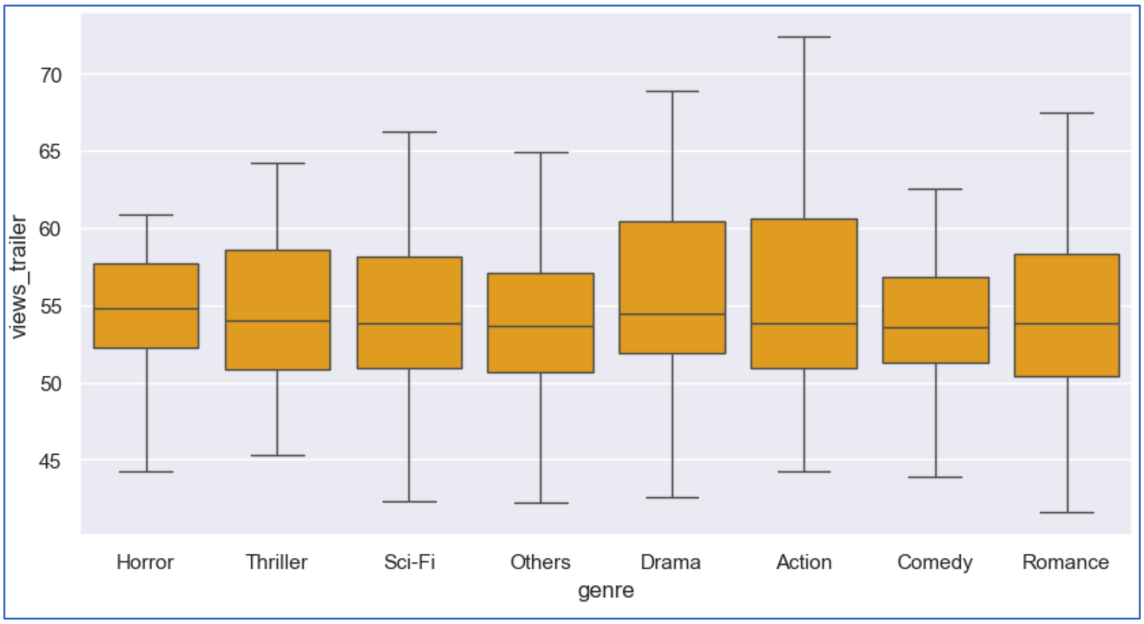


Fig 36. views_trailer v/s genre – Box Plot

OBSERVATIONS:

- Horror and drama content receive the highest number of trailer views among all genres, while comedy content garners the lowest.

4.9 Insights based on Exploratory Data Analysis

Based on Univariate analysis:

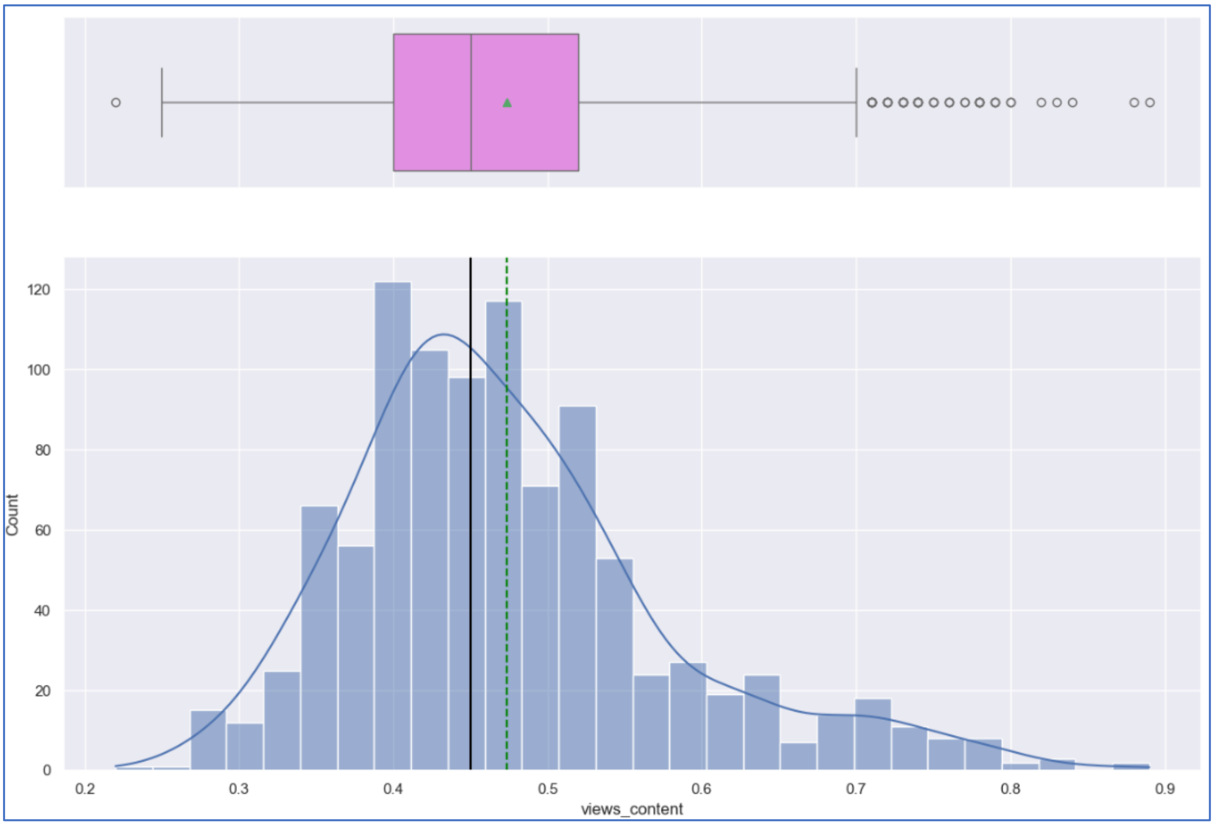
- On an average, 1.7 million people visit the OTT platform in a week, out of which the content trailer attracts viewership ranging from 0.2 million to 1 million, with an average viewership of 0.49 million.
- The trailer views range from 25 million to 200 million, with a concentration of observations between 40 million and 65 million.
- The average number of ad impressions is found to be 1450 million.
- Major sports event is absent 60% of the time.
- Genres are distributed nearly equally, with a slight uptick in comedy content.
- The release of content peaks during winter, particularly on Fridays.

Based on Bivariate analysis:

- The content views on the first day are significantly influenced by both the trailer views and the number of visitors to the OTT platform.
- Content views peak when major sports events are not taking place and are notably higher on weekends, specifically Saturdays and Sundays.
- The impact of platform visitors and trailer views on first-day content views remains consistent across all genres, seasons and days of the week.
- Ad impressions do not significantly contribute to the data.
- Winter and summer are optimal seasons for releasing content on the OTT platform.

5. Questions to be answered as a part of the EDA section

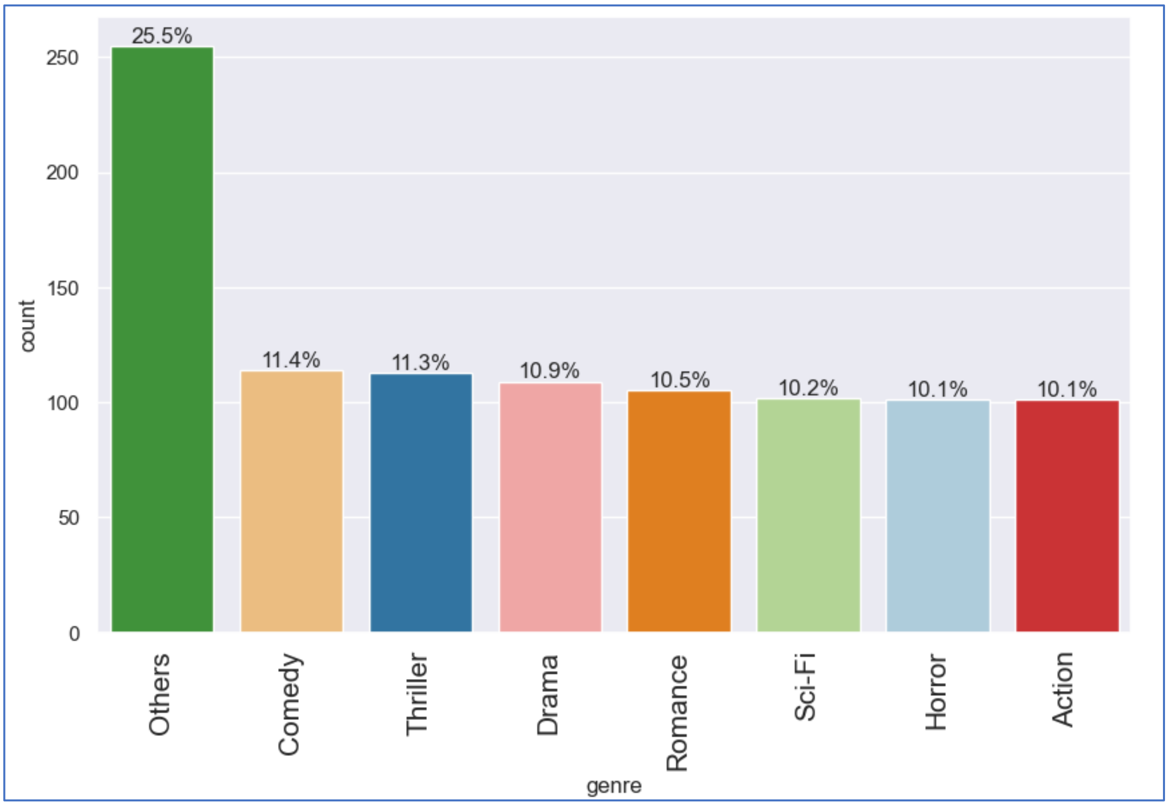
5.1. What does the distribution of content views look like?



ANSWER:

- The content views are normally distributed with a right-skewed distribution.
- The mean and median are almost close to each other, where the average being 0.49 million views on the first-day of release.
- Few of the contents are closer to 1 million views.
- The content views span over a range of 0.2 million to 1 million.

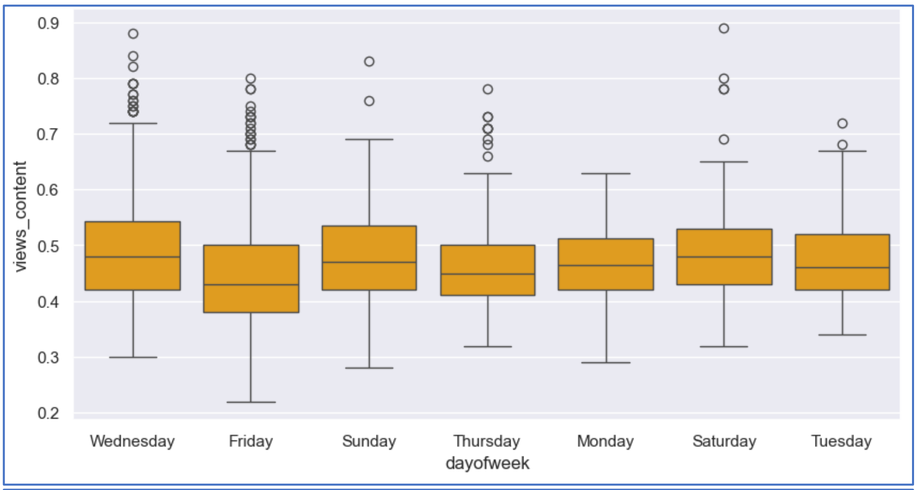
5.2 What does the distribution of genres look like?

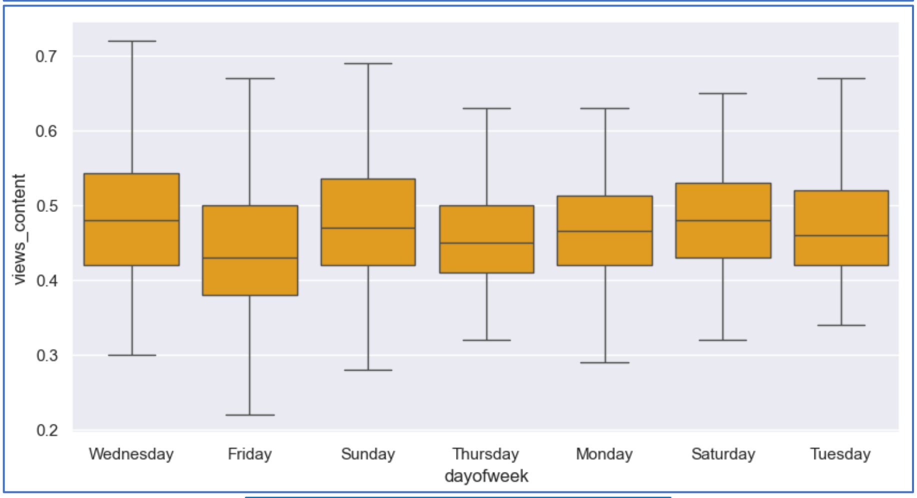


ANSWER:

- Almost all the genres are equally distributed in the OTT platform ranging from 10.1% for action (low) to 11.4% for comedy (high).
- Apart from the mentioned genres, the remaining category accounts to 25.5%.

5.3 The day of the week on which content is released generally plays a key role in the viewership. How does the viewership vary with the day of release?





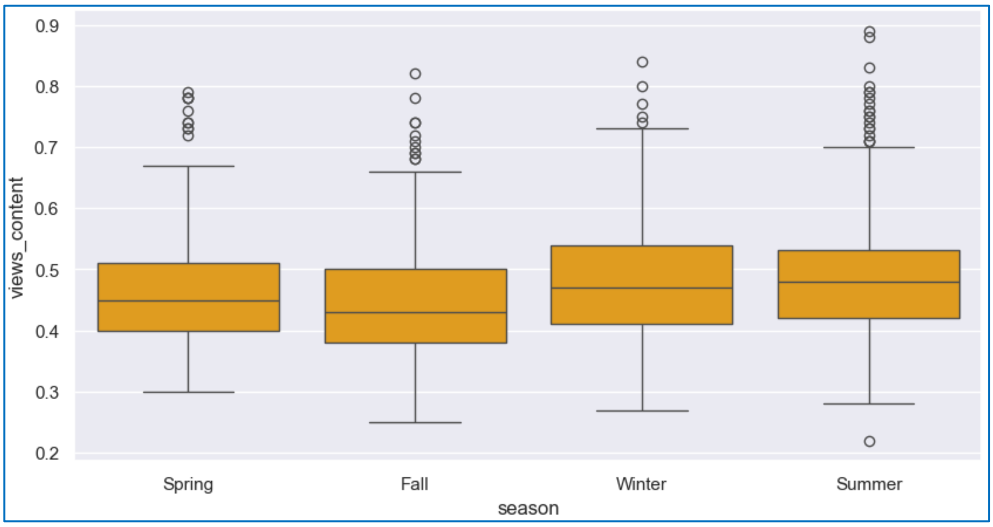
dayofweek	views_content
Friday	0.446694
Monday	0.467917
Saturday	0.497955
Sunday	0.484179
Thursday	0.470619
Tuesday	0.487826
Wednesday	0.494608

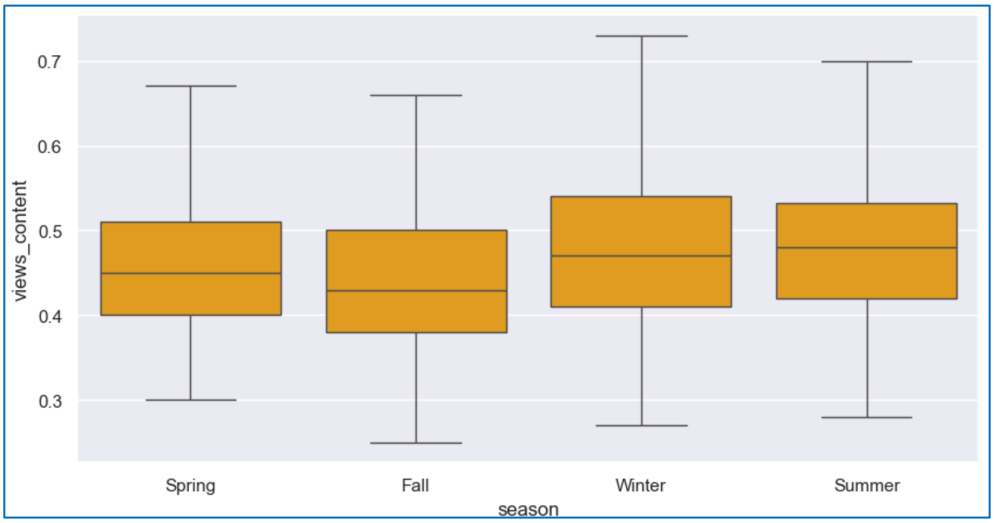
ANSWER:

- On an average, the contents released on Saturdays, Wednesdays, Tuesdays and Sundays, have high number of views, in millions, on the first-day of content release.
- The contents released on Fridays, Mondays and Thursdays have low number of views, in millions, on the first-day of content release comparatively.

5.4 How does the viewership vary with the season of release?

CONTENT VIEWERSHIP AND SEASON:



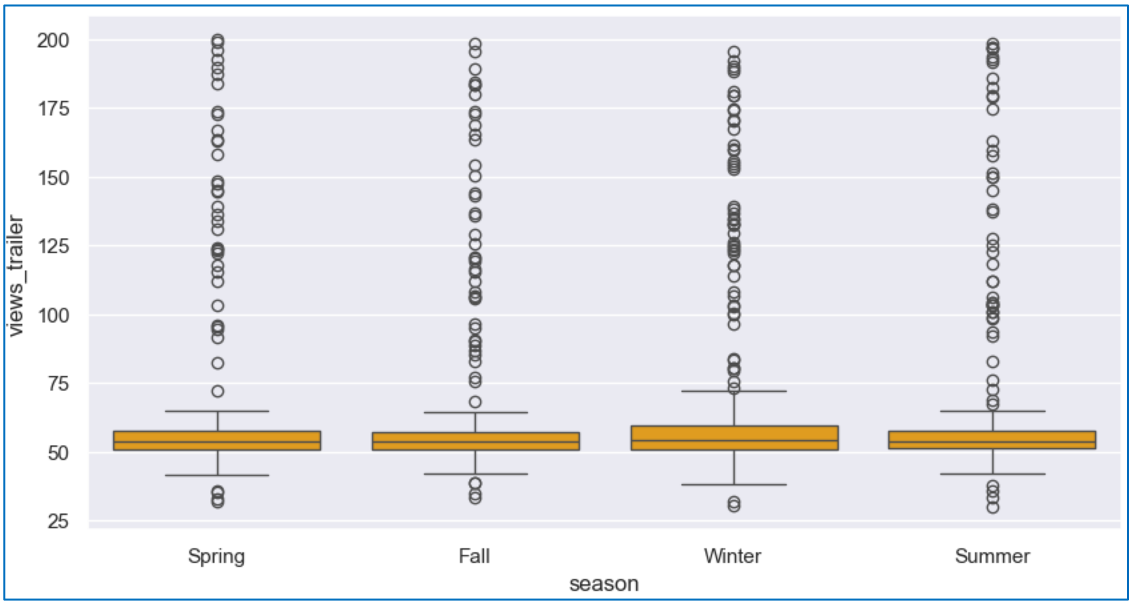


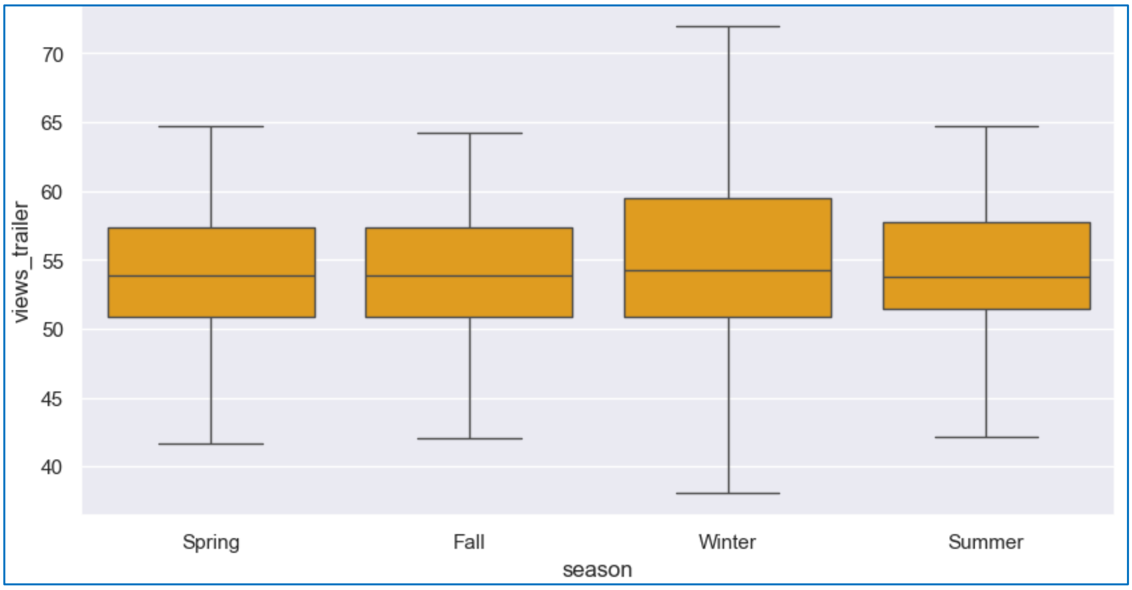
season	views_content
Fall	0.445357
Spring	0.467166
Summer	0.496803
Winter	0.484669

ANSWER

- Content released during Summer receives a significant number of views on its first day of release, followed by Winter.
- Content released in Spring and Fall receive relatively fewer views on their first day of release.

TRAILER VIEWERSHIP AND SEASON



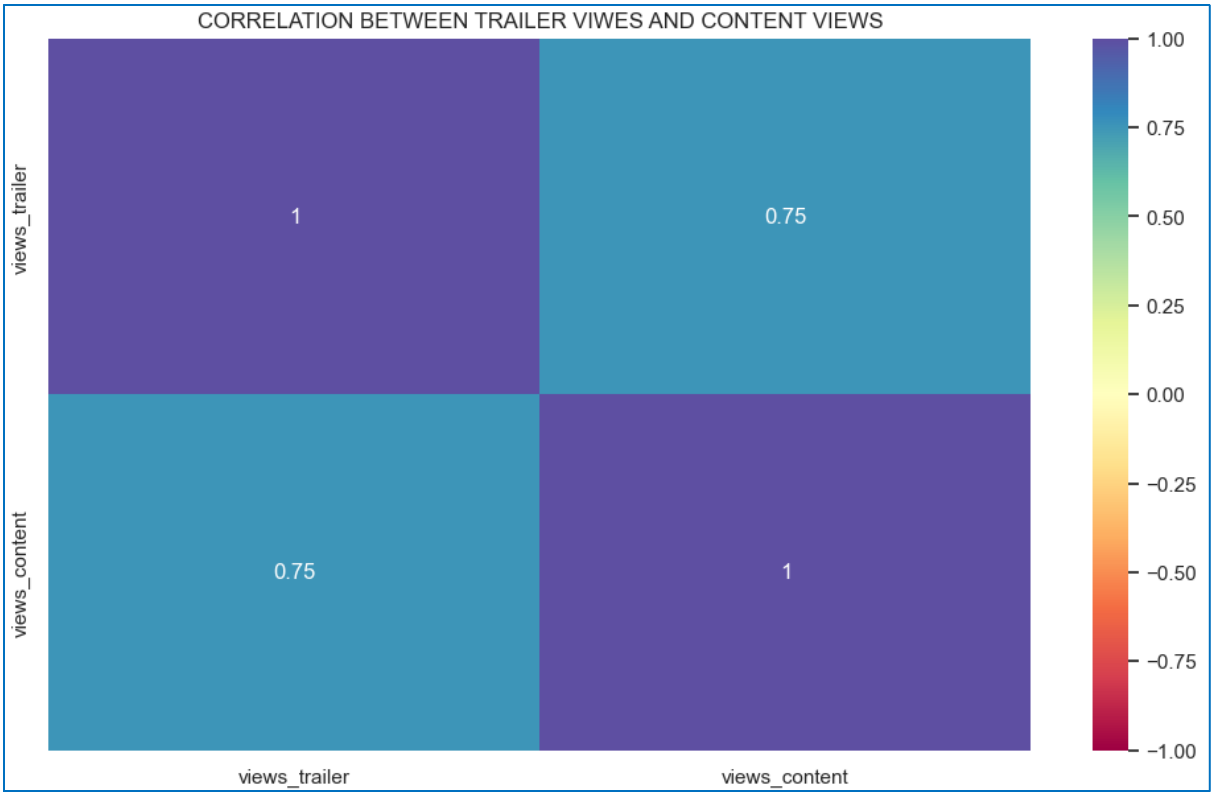


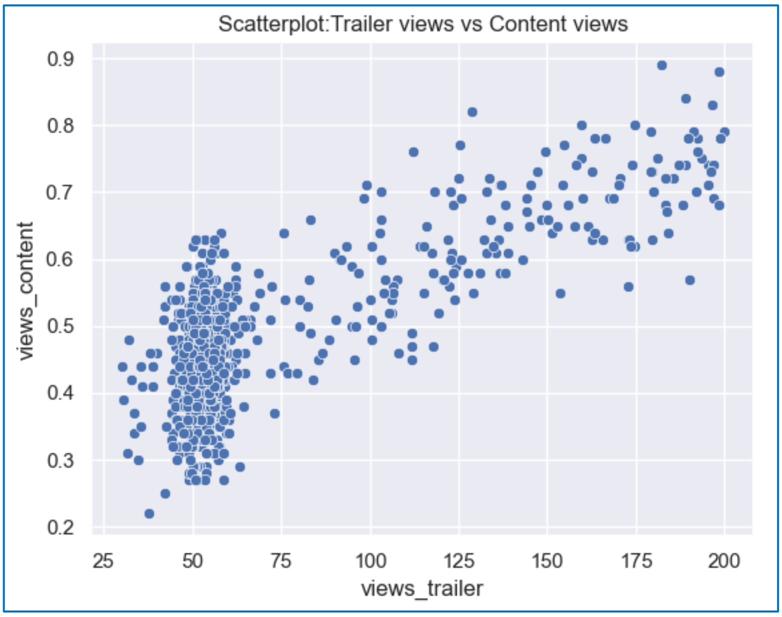
	views_trailer
season	
Fall	65.339127
Spring	65.247571
Summer	67.420861
Winter	69.584786

ANSWER

- Trailer released during Winter receives a significant number of views on its first day of release, followed by Summer.
- Trailer released in Spring and Fall receive relatively fewer views on their first day of release.

5.5 What is the correlation between trailer views and content views?





ANSWER

- There is high correlation, about 0.75 between trailer views and content views.
- The relationship between trailer views and content views is linear. As the trailer views increases, the content views also increase.

6. Data Preprocessing

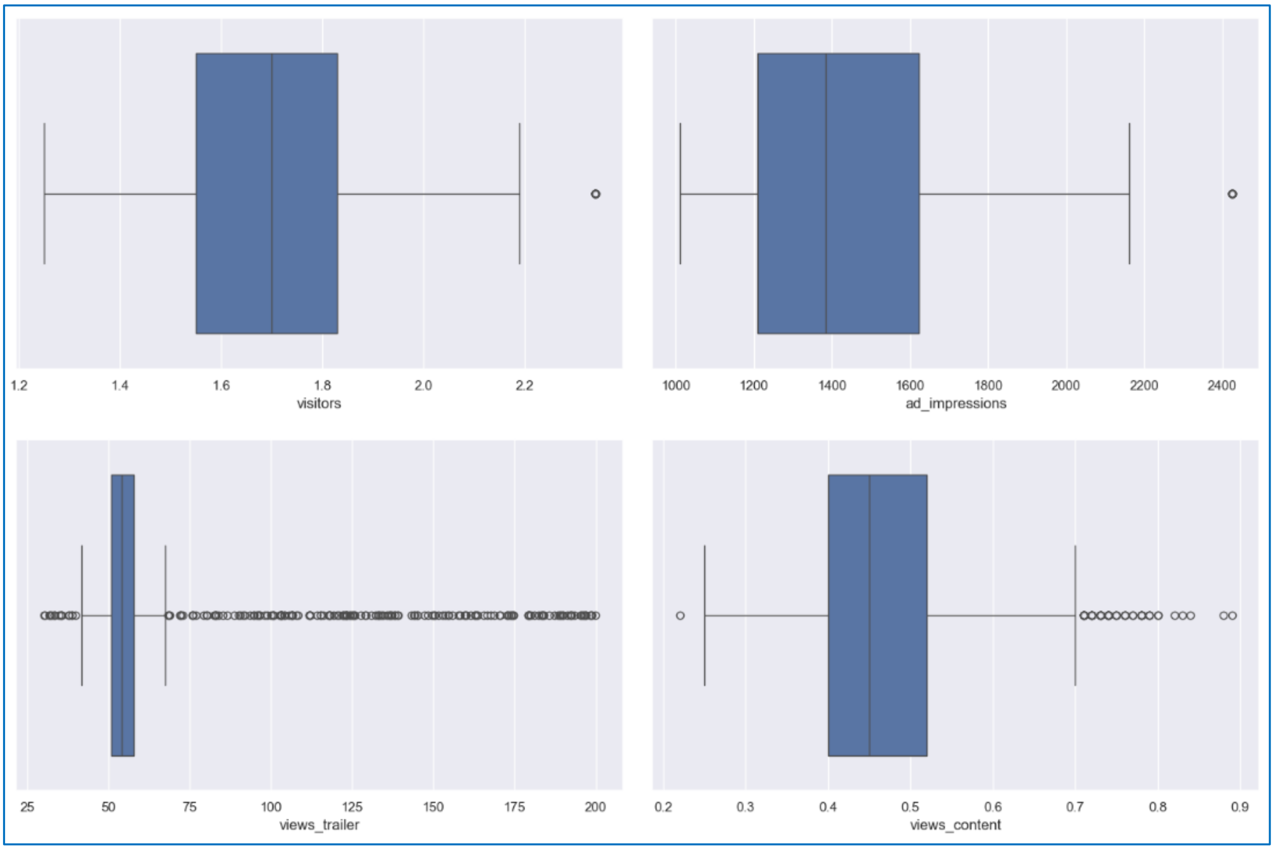
6.1 Duplicate value check

- No duplicate values are found in the data set.

6.2 Missing value treatment

- No missing values or null values are found in the data set.

6.3 Outlier detection



- There are a few outliers in the data.
- However, we will not treat them as they are proper values which adds value to the data.

6.4 Feature engineering

- There are 4 categorical variables in the data set, where major_sports_event in of int data type.
- The values 0 and 1, which is the binary representation for the absence and presence of major sports event in a day has been converted to No and Yes respectively.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   visitors            1000 non-null   float64
 1   ad_impressions      1000 non-null   float64
 2   genre               1000 non-null   object 
 3   dayofweek           1000 non-null   object 
 4   season              1000 non-null   object 
 5   views_trailer       1000 non-null   float64
 6   views_content       1000 non-null   float64
 7   Major_sports_event  1000 non-null   object 
dtypes: float64(4), object(4)
memory usage: 62.6+ KB
```

6.5 Data Preparation for Modelling

The steps involved in data preparation for modelling include:

- Encoding the categorical features in order to predict the factors influencing the first-day views of the content on the OTT platform before building a model.
- Adding intercept in order to explain the linear equation y=a+bx
- Creating dummy variables for the categorical variables in the data set, which includes major_sports_event, dayofweek, genre and season.

genre_Comedy	genre_Drama	genre_Horror	genre_Others	genre_Romance	genre_Sci-Fi	...	dayofweek_Monday	dayofweek_Saturday	dayofweek_Sunday
0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0	0.0
0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0	1.0

- Splitting the data into train and test, where the train constitutes 70% and test constitutes 30%.

```
Number of rows in train data = 700
Number of rows in test data = 300
```

7. Model Building- Linear Regression

7.1 Building Linear Regression Model using the train data

- The linear regression model is built using the ordinary least squares (OLS) model of regression.

7.2 Regression results:

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.792
Model:                            OLS   Adj. R-squared:                  0.785
Method:                 Least Squares   F-statistic:                     129.0
Date:                Fri, 05 Jul 2024   Prob (F-statistic):          1.32e-215
Time:                        17:09:51   Log-Likelihood:                 1124.6
No. Observations:                 700   AIC:                            -2207.
Df Residuals:                     679   BIC:                            -2112.
Df Model:                          20
Covariance Type:            nonrobust
========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------
const                    0.0602      0.019      3.235      0.001       0.024       0.097
visitors                 0.1295      0.008     16.398      0.000       0.114       0.145
ad_impressions        3.623e-06   6.58e-06      0.551      0.582    -9.3e-06    1.65e-05
views_trailer            0.0023   5.52e-05     42.193      0.000       0.002       0.002
genre_Comedy             0.0094      0.008      1.172      0.241      -0.006       0.025
genre_Drama              0.0126      0.008      1.554      0.121      -0.003       0.029
genre_Horror             0.0099      0.008      1.207      0.228      -0.006       0.026
genre_Others             0.0063      0.007      0.897      0.370      -0.008       0.020
genre_Romance            0.0006      0.008      0.065      0.948      -0.016       0.017
genre_Sci-Fi             0.0131      0.008      1.599      0.110      -0.003       0.029
genre_Thriller           0.0087      0.008      1.079      0.281      -0.007       0.025
dayofweek_Monday         0.0337      0.012      2.848      0.005       0.010       0.057
dayofweek_Saturday       0.0579      0.007      8.094      0.000       0.044       0.072
dayofweek_Sunday         0.0363      0.008      4.639      0.000       0.021       0.052
dayofweek_Thursday       0.0173      0.007      2.558      0.011       0.004       0.031
dayofweek_Tuesday        0.0228      0.014      1.665      0.096      -0.004       0.050
dayofweek_Wednesday      0.0474      0.004     10.549      0.000       0.039       0.056
season_Spring            0.0226      0.005      4.224      0.000       0.012       0.033
season_Summer            0.0442      0.005      8.111      0.000       0.034       0.055
season_Winter            0.0272      0.005      5.096      0.000       0.017       0.038
Major_sports_event_Yes  -0.0603      0.004    -15.284      0.000      -0.068      -0.053
==============================================================================
Omnibus:                        3.850   Durbin-Watson:                   2.004
Prob(Omnibus):                  0.146   Jarque-Bera (JB):                3.722
Skew:                           0.143   Prob(JB):                        0.156
Kurtosis:                       3.215   Cond. No.                     1.67e+04
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 1.67e+04. This might indicate that there are
strong multicollinearity or other numerical problems.
```

- Adjusted. R-squared is 0.785 which is a higher value and indicates a better fit, assuming certain conditions are met.
- const coefficient is 0.0602 which is the Y-intercept. The const coefficient indicates that if all the predictor variable coefficients are zero, then the expected output would be equal to the const coefficient.

7.3 Model coefficients with column names

```
                              coef
------------------------------------
const                       0.0602
visitors                    0.1295
ad_impressions           3.623e-06
views_trailer               0.0023
genre_Comedy                0.0094
genre_Drama                 0.0126
genre_Horror                0.0099
genre_Others                0.0063
genre_Romance               0.0006
genre_Sci-Fi                0.0131
genre_Thriller              0.0087
dayofweek_Monday            0.0337
dayofweek_Saturday          0.0579
dayofweek_Sunday            0.0363
dayofweek_Thursday          0.0173
dayofweek_Tuesday           0.0228
dayofweek_Wednesday         0.0474
season_Spring               0.0226
season_Summer               0.0442
season_Winter               0.0272
Major_sports_event_Yes     -0.0603
```

7.5 Model Performance Evaluation

The performance of the model on the training dataset is evaluated using metrics such as root mean square error (RMSE), adjusted R-squared, mean absolute error (MAE) and mean absolute percentage error (MAPE).

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.04853	0.038197	0.791616	0.785162	8.55644

Observations:

- The training data set has an adjusted R-squared value of 0.785, so the model is not underfitting.
- The train and test RMSE and MAE are comparable, so the model is not overfitting.
- MAE suggests that the model can predict content views on the first-day of release within a mean error of 0.038 on the test data.
- MAPE of 8.55 on the test data means that we are able to predict within 8.55% of the content views on the first-day of release.

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8. Checking Linear Regression Assumptions

8.1 Test for multicollinearity

- Multicollinearity is a situation where the predictions in a regression model are linearly dependent.
- It is essential for the dataset to be free of multicollinearity to ensure the development of a robust model.

8.1.1 Variation inflation factor (VIF)

- Multicollinearity is tested using VIF which measures the inflation in the variances of the regression parameter estimates due to collinearities that exist among the predictors.
- If VIF is 1, then there is no correlation among the kth predictor and the remaining predictor variables.

	feature	VIF
0	const	99.679317
1	visitors	1.027837
2	ad_impressions	1.029390
3	views_trailer	1.023551
4	genre_Comedy	1.917635
5	genre_Drama	1.926699
6	genre_Horror	1.904460
7	genre_Others	2.573779
8	genre_Romance	1.753525
9	genre_Sci-Fi	1.863473
10	genre_Thriller	1.921001
11	dayofweek_Monday	1.063551
12	dayofweek_Saturday	1.155744
13	dayofweek_Sunday	1.150409
14	dayofweek_Thursday	1.169870
15	dayofweek_Tuesday	1.062793
16	dayofweek_Wednesday	1.315231
17	season_Spring	1.541591
18	season_Summer	1.568240
19	season_Winter	1.570338
20	Major_sports_event_Yes	1.065689

Observations

- The VIF value for the above regression model falls between 1-2, which indicates that there is too low or no correlation among the predictor and the remaining predictor variables, and hence, the variance of βk is not inflated.

8.1.2 Dealing with high p-value variables

- Few of the dummy variables in the data set have p-value > 0.05. They are not significant and hence need to be dropped.
- The p-values change after dropping a variable. So, the variables are not dropped all at once.
- Instead, a model is built and the p-values of the variables are checked and column with highest p-value is dropped.
- A new model is created that includes all features, p-values of the variables are assessed and the column with the highest p-value is eliminated.
- The above two steps are repeated till there are no columns with p-value > 0.05.
- To perform the above steps, using loop will be more efficient.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.786
Method:                 Least Squares   F-statistic:                     233.8
Date:                Sat, 06 Jul 2024   Prob (F-statistic):          7.03e-224
Time:                        21:41:12   Log-Likelihood:                 1120.2
No. Observations:                 700   AIC:                            -2216.
Df Residuals:                     688   BIC:                            -2162.
Df Model:                          11
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                       0.0747      0.015      5.110      0.000       0.046       0.103
visitors                    0.1291      0.008     16.440      0.000       0.114       0.145
views_trailer               0.0023    5.5e-05     42.414      0.000       0.002       0.002
dayofweek_Monday            0.0321      0.012      2.731      0.006       0.009       0.055
dayofweek_Saturday          0.0570      0.007      8.042      0.000       0.043       0.071
dayofweek_Sunday            0.0344      0.008      4.456      0.000       0.019       0.050
dayofweek_Thursday          0.0154      0.007      2.307      0.021       0.002       0.029
dayofweek_Wednesday         0.0465      0.004     10.532      0.000       0.038       0.055
season_Spring               0.0226      0.005      4.259      0.000       0.012       0.033
season_Summer               0.0434      0.005      8.112      0.000       0.033       0.054
season_Winter               0.0282      0.005      5.362      0.000       0.018       0.039
Major_sports_event_Yes     -0.0606      0.004    -15.611      0.000      -0.068      -0.053
==============================================================================
Omnibus:                        3.254   Durbin-Watson:                   1.996
Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Observations

- No variable in the data set has p-value greater than 0.05, hence we can consider the variables in the x_train1 as the final set of predictor variables and olsmod1 as the final model to move forward.
- The final adjusted R-squared is 0.785, which suggests that the model explains approximately 78.5% of the variance.

8.1.3 Assessing train data and test data performance

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.048841	0.038385	0.788937	0.785251	8.595246

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

Observations

- The adjusted R-squared in olsmod was 0.785 and the adjusted R-squared in olsmod1 is 0.786, which indicates that the variables we dropped were not affecting the model to a greater extent.
- RMSE and MAE values are comparable for train and test sets, indicating that the model is not overfitting.

The assumption of multicollinearity is satisfied.

8.2 Test for linearity and independence

- Linearity describes a straight-line relationship between two variables, where the predictor variables must have a linear relation with the dependent variable.
- A plot showing the relationship between fitted values and residuals is generated.
- If they don't follow any pattern, then the model is linear and residuals are independent.
- Otherwise, the model shows signs of non-linearity and residuals are not independent.

	Actual Values	Fitted Values	Residuals
731	0.40	0.445434	-0.045434
716	0.70	0.677403	0.022597
640	0.42	0.433999	-0.013999
804	0.55	0.562030	-0.012030
737	0.59	0.547786	0.042214

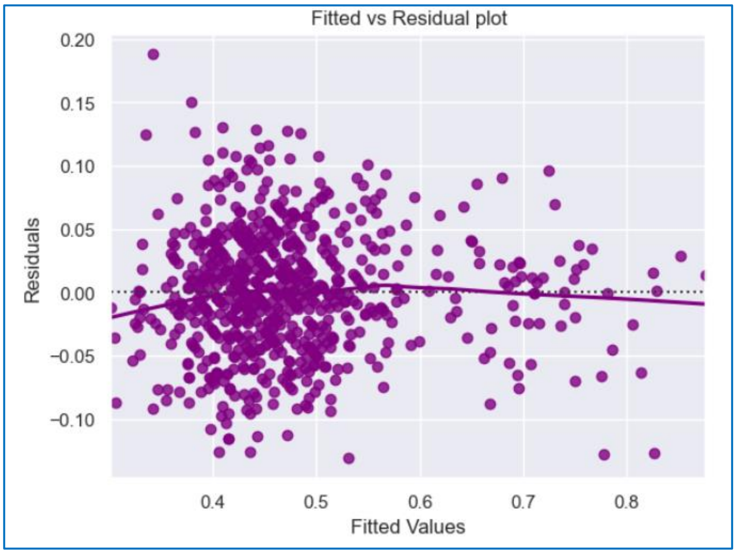

Fig 37. Fitted vs Residual Plot

Observations:

- There is no pattern seen in the above scatterplot plotted for residuals and fitted values.

The assumptions of linearity and independence are satisfied.

8.3 Test for normality

- The residuals should be normally distributed.
- If the error terms are not normally distributed, confidence intervals of the coefficient estimates may become too wide or narrow.
- Once confidence interval becomes unstable, it leads to difficulty in estimating coefficients based on minimization of least squares.
- Non-normality suggests that there are a few unusual data points that must be studied closely to make a better model.

8.3.1 Histogram

- The shape of the histogram of residuals can give an initial idea about the normality.

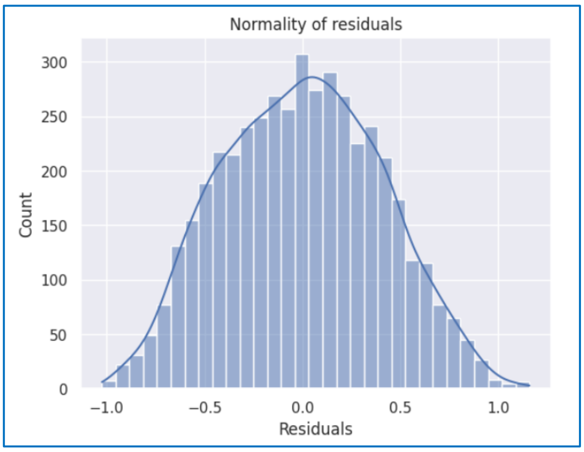

Fig 38. Normality of Residuals Plot

OBSERVATIONS:

- The histogram of residuals has a bell shape.

8.3.2 Q-Q plot

- If the residuals follow a normal distribution, they will make a straight-line plot, otherwise not.

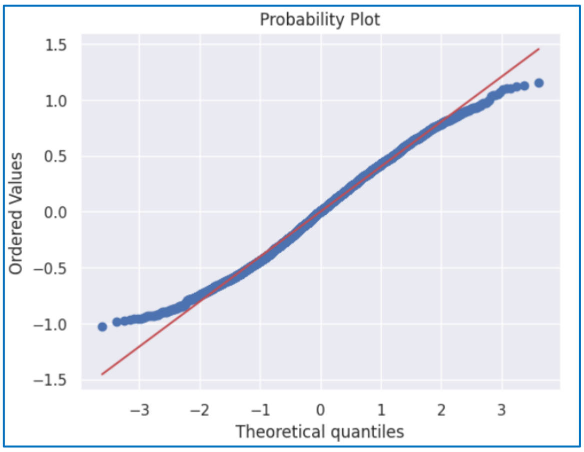

Fig 39. Q-Q Plot

OBSERVATIONS:

- The residuals more or less follow a straight line except for the tails.

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8.3.4 Shapiro-Wilk test

- Null hypothesis: Residuals are normally distributed
- Alternate hypothesis: Residuals are not normally distributed

```
ShapiroResult(statistic=0.9934884905815125, pvalue=1.4591795924768364e-13)
```

OBSERVATIONS:

- Since p-value is higher than 0.05, the residuals are normal as per the Shapiro-Wilk test.

The assumption of normality is satisfied.

8.4 Test for homoscedasticity

- If the variance of the residuals is symmetrically distributed across the regression line, then the data is said to be homoscedastic.

8.4.1 Goldfeldquandt test

- Null hypothesis: Residuals are homoscedastic.
- Alternate hypothesis: Residuals have heteroscedasticity.

```
[('F statistic', 1.131361290420075), ('p-value', 0.12853551819087372)]
```

OBSERVATIONS:

- Since p-value is higher than 0.05, the residuals are homoscedastic.

The assumption of homoscedasticity is satisfied.

8.5 Predictions on test data

	Actual	Predicted
983	0.43	0.434802
194	0.51	0.500314
314	0.48	0.430257
429	0.41	0.492544
267	0.41	0.487034
746	0.68	0.680000
186	0.62	0.595078
964	0.48	0.503909
676	0.42	0.490313
320	0.58	0.560155

OBSERVATIONS:

- We can observe here that the model has returned pretty good prediction results, and the actual and predicted values are comparable.

8.6 Final Model

```
                            OLS Regression Results
==============================================================================
Dep. Variable:          views_content   R-squared:                       0.789
Model:                            OLS   Adj. R-squared:                  0.786
Method:                 Least Squares   F-statistic:                     233.8
Date:                Sat, 06 Jul 2024   Prob (F-statistic):          7.03e-224
Time:                        22:21:18   Log-Likelihood:                 1120.2
No. Observations:                 700   AIC:                            -2216.
Df Residuals:                     688   BIC:                            -2162.
Df Model:                          11
Covariance Type:            nonrobust
========================================================================================
                           coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------------
const                    0.0747      0.015      5.110      0.000       0.046       0.103
visitors                 0.1291      0.008     16.440      0.000       0.114       0.145
views_trailer            0.0023    5.5e-05     42.414      0.000       0.002       0.002
dayofweek_Monday         0.0321      0.012      2.731      0.006       0.009       0.055
dayofweek_Saturday       0.0570      0.007      8.042      0.000       0.043       0.071
dayofweek_Sunday         0.0344      0.008      4.456      0.000       0.019       0.050
dayofweek_Thursday       0.0154      0.007      2.307      0.021       0.002       0.029
dayofweek_Wednesday      0.0465      0.004     10.532      0.000       0.038       0.055
season_Spring            0.0226      0.005      4.259      0.000       0.012       0.033
season_Summer            0.0434      0.005      8.112      0.000       0.033       0.054
season_Winter            0.0282      0.005      5.362      0.000       0.018       0.039
Major_sports_event_Yes  -0.0606      0.004    -15.611      0.000      -0.068      -0.053
==============================================================================
Omnibus:                        3.254   Durbin-Watson:                   1.996
Prob(Omnibus):                  0.196   Jarque-Bera (JB):                3.077
Skew:                           0.139   Prob(JB):                        0.215
Kurtosis:                       3.168   Cond. No.                         662.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Training Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.048841	0.038385	0.788937	0.785251	8.595246

Test Performance

	RMSE	MAE	R-squared	Adj. R-squared	MAPE
0	0.051109	0.041299	0.761753	0.751792	9.177097

OBSERVATIONS:

- The model is able to explain approximately 75% of the variation in the data.
- The train and test RMSE and MAE are low and comparable. So, the model is not suffering from overfitting.
- The MAPE on the test set suggests we can predict within 9.1% of the content views on the first-day of release.
- Hence, we can conclude the model olsmodel_final is good for prediction as well as inference purposes.

9. Equation of linear regression

The equation of linear regression obtained from the ShowTime data set is calculated as follows:

views_content = 0.07467052053720907 + 0.1290958182589414 * (visitors) + 0.0023308167861640196 * (views_trailer) + 0.03206580679023644 * (dayofweek_Monday) + 0.05702859660165121 * (dayofweek_Saturday) + 0.03438622992362496 * (dayofweek_Sunday) + 0.015449441769973302 * (dayofweek_Thursday) + 0.046494803669848137 * (dayofweek_Wednesday) + 0.022604915818118076 * (season_Spring) + 0.04339100263609975 * (season_Summer) + 0.028230557183976834 * (season_Winter) + -0.06055507818137332 * (Major_sports_event_Yes)

10. Actionable Insights & Recommendations

- The model is able to explain approximately 75% of the variation in the data and can predict 9.1% of the content views on the first-day of release on the test data, which indicates that the model is good for prediction as well as inference purposes.
- If the visitors of the OTT platform increase by one unit, then the content views on the first-day of release increases by 0.1291 units, all other variables held constant.
- If the number of trailer views increases by one unit, then the content views on the first-day of release increases by 0.0023 units, all other variables held constant.
- If the presence of major sports event increases by one unit, then the content views on the first-day of release decreases by 0.06 units, all other variables held constant.
- The content views on the first-day of release on a Saturday will be 0.05 units more than those released on a Friday.
- As the content views on the first-day of release increases with the increase in the trailer views, the company can increase the promotion on the trailer content.
- As the content views are higher in summers and winters, the ShowTime platform can bring new contents to the OTT during this time.
- Ad impressions do not contribute to the value of number of views obtained for the content, thus it is required to enhance either the content itself or the timing of the ads.
- Since Sci-Fi, Horror, and Action genres receive the highest number of views on the first day of release, it would be beneficial to prioritize content in these categories on the platform.
- The timing of higher viewership on the OTT platform can impact the selection of engaging content across genres, hence it is advisable to collect data on these timings.
- Increasing the number of visitors on the OTT platform positively impacts content views. Therefore, attracting more users can be achieved by reducing subscription prices or offering additional benefits.
- Gathering data on the age and gender of platform visitors would aid in understanding the viewer demographics, facilitating deeper analysis.